

**2025 MANAGEMENT PLAN
FOR
COMMERCIAL SHELLFISH GROWING AREA
LEASE M-653-02
IN
SANTA BARBARA, CALIFORNIA**

California Department of Public Health
Division of Radiation Safety and Environmental Management
Environmental Management Branch
Environmental Health Services Section
Preharvest Shellfish Unit

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II. ABBREVIATIONS AND DEFINITIONS

This document contains many acronyms and abbreviations. In general, an abbreviation will be given in parentheses () following the first time a title or term is used, and the abbreviation will be used in almost all cases in place of that term later. The following alphabetical list of abbreviations used in this document is provided to assist the reader:

<u>ABBREVIATION</u>	<u>DESCRIPTION</u>
Cal-OES	California Governor's Office of Emergency Services
CCRWQCB	California Central Coast Regional Water Quality Control Board
CDFW	California Department of Fish and Wildlife
CDPH	California Department of Public Health
CDPH/FDB	Food and Drug Branch (within CDPH)
CDPH/PSU	California Department of Public Health, Preharvest Shellfish Unit
FDA	U.S. Food and Drug Administration
EMB	Environmental Management Branch (within CDPH)
GIL	Growers' Information Line
H & S §	Section of the California Health and Safety Code
ICSSL	Interstate Certified Shellfish Shippers List
LEO	Laboratory Evaluation Officer
MGD	Million Gallons per Day
mL	Milliliters
MOU	Memorandum of Understanding
MPN	Most Probable Number
NSSP	National Shellfish Sanitation Program
PSP	Paralytic Shellfish Poisoning
RDO	Rainfall Duty Officer (a designated staff person in CDPH/PSU)
SBCEHS	Santa Barbara County Environmental Health Services
SBMC	Santa Barbara Mariculture Company
SOP	Standard Operating Procedures
SRS	Systematic Random Sampling
SSP	State Shellfish Program
WDRs	Waste Discharge Requirements (adopted by CCRWQCB)
WWTP	Waste Water Treatment Plant

The following definitions shall apply to this document:

Certified Laboratory. A laboratory that has been evaluated by the proper authority for a specified laboratory method and found to be in conformance with the National Shellfish Sanitation Program requirements.

Direct marketing. Direct marketing means the sale for human consumption of shellfish which does not require depuration or relaying prior to sale.

Harvest. The act of removing shellstock from growing areas and its placement on or in a manmade conveyance or other means of transport.

Harvest time. For the purpose of time to temperature control, harvest time begins once the first shellstock harvested is no longer submerged.

Seed. "Seed" means shellstock which is less than market size.
To ensure consistency amongst west coast aquaculture and to provide guidance to the certified grower in the Santa Barbara Channel, the following seed size definitions are provided.

Mussels	1/2 inch (13 millimeters)
Scallops	1 inch (25 millimeters)
Kumamoto oysters	1 inch (25 millimeters)
Olympia oysters	3/4 inch (19 millimeters)
Pacific oysters	1 1/2 inch (38 millimeters)
Other oyster species	1 1/2 inch (38 millimeters)
Clam species	1/2 inch (13 millimeters)

Shall. The term "shall" is used in this plan in its legal or regulatory sense to denote a mandatory requirement.

Shellfish grower or harvester. For the purposes of this plan, the terms "grower" and "harvester" refer to the same persons or firms and may be used interchangeably.

Wet storage. Wet storage means the storage, by a dealer, of shellstock from growing areas in the approved classification or in the open status of the conditionally approved classification in containers or floats in natural bodies of water or in tanks containing natural or synthetic seawater at any permitted land-based activity or facility.

III. INTRODUCTION

This Management Plan establishes the standards and procedures used to regulate commercial shellfish harvesting at commercial aquaculture Lease M-653-02 and has been prepared for adoption in accordance with the procedures set forth in Sanitary Control of Shellfish, (H & S § 112150 Division 104, Part 6, Chapter 5) and will constitute an order of the Director of the California Department of Public Health (CDPH) as described in H & S § 112160(c).

Commercial shellfish aquaculture Lease M-653-02 is located in the Santa Barbara Channel, approximately 3200 feet from the shore at its nearest point off of Arroyo Burro Beach in Santa Barbara County. This shellfish growing area is approximately 72 acres, 4000 feet long by 880 feet wide, with the length running approximately parallel to the shore (Figure 1). The California Department of Fish and Wildlife (CDFW) has issued this aquaculture lease to Mr. Bernard Friedman, owner and operator of the Santa Barbara Mariculture Company (SBMC), for the purpose of growing and harvesting oysters, clams, mussels, and scallops. Non-point sources of contamination from the watershed for this area have the potential for impacting the sanitary quality of the growing area, mainly during rainfall related runoff from local drainages. There are also two municipal wastewater treatment plants (WWTPs) that discharge effluent through outfalls in the Santa Barbara Channel and the CDPH calculated zone of influence for the outfall reaches the SBMC lease. These WWTPs include the City of Santa Barbara's El Estero Wastewater Treatment Plant, and the collection, treatment, and disposal system of the Goleta Sanitary District. Spills or malfunctions of the disinfection process could impact the growing area. The WWTPs of Montecito, Summerland, and Carpinteria also discharge into the Santa Barbara Channel but have CDPH calculated zones of influence that do not reach the SBMC lease and are therefore not included in this plan.

Because Lease M-653-02 may be subject to intermittent microbiological pollution from various sources, the commercial shellfish growing area is classified as *Conditionally Approved* for commercial harvest. The National Shellfish Sanitation Program (NSSP) Model Ordinance (2023), Chapter IV, requires development of a Management Plan for any *Conditionally Approved* growing area. This Management Plan sets forth the standards and criteria necessary to manage the harvesting of bivalve molluscan shellfish intended for human consumption.

The standards, conditions, and procedures for closure of the growing area at times of predictable elevated pollution following periods of significant rainfall, as well as for emergency closures established for unpredictable events such as sewage spills or oil spills have been set forth in this Management Plan such that when the specified events occur, predetermined actions are implemented. Any temporary closure to shellfish harvesting resulting from these pollution events that may be declared pursuant to this Management Plan shall be deemed to meet the requirements of Section 112160(d) pertaining to notice and opportunity to submit data, views, or arguments. Rainfall closures declared under this Management Plan are not considered "emergency actions" referred to in H & S § 112160(e), but instead are dealt with as part of the Director's order establishing this plan. Compliance by the certified shellfish harvester with the

conditions and procedures set forth in this Management Plan is mandatory as a condition of the Shellfish Growing Area Certificate issued for the aquaculture growing area.

It is a violation of state laws and regulations to culture, harvest, or hold shellfish intended for sale for human food except in or from areas that have been specifically identified in a valid Shellfish Growing Area Certificate issued by CDPH/PSU. The holding of shellfish in any other waters prior to on-shore processing, storage, and marketing is a prohibited activity. Any shellfish held in uncertified waters, or in water-filled tanks or containers, which have not been specifically approved by CDPH/PSU for that purpose, are subject to embargo and/or destruction by CDPH/PSU or its authorized agents.

Significant changes to the Management Plan for Lease M-653-02 can be found in Appendix G.

IV. ORGANIZATION AND RESPONSIBILITIES

Numerous agencies and individuals are involved parties to the management of the Lease M-653-02 Aquaculture Area. A contact list is provided in Appendix A.

A. CALIFORNIA DEPARTMENT OF PUBLIC HEALTH

California Department of Public Health (CDPH) is the lead agency for the State Shellfish Program (SSP) and is the agency responsible for growing area classification, issuing growing area certificates and for decision making regarding harvest closures and reopening of commercial shellfish growing areas. Within CDPH, the Deputy Director of the Center for Environmental Health has overall responsibility for the California SSP. Two major program segments are housed in the following branches:

1. Environmental Management Branch

The Chief of the Environmental Management Branch (EMB), through the Chief of the Environmental Health Services Section, is responsible for the CDPH Preharvest Shellfish Unit (CDPH/PSU). The Chief of the CDPH/PSU heads a staff of several Environmental Scientists and coordinates activities as it relates to the evaluation, classification, and certification of commercial shellfish growing areas as well as routine monitoring for marine biotoxins.

2. Food and Drug Branch

The Chief of the Food and Drug Branch (CDPH/FDB) is responsible for the postharvest shellfish program as it relates to the regulation of shellfish handling, processing, storage, and distribution after harvest, including the labeling and identification of shellfish moving in commerce. FDB provides assistance in the area of patrol, prevention of illegal shellfish harvesting and illness investigations.

B. U.S. FOOD AND DRUG ADMINISTRATION

A primary responsibility of the U.S. Food and Drug Administration (FDA) is to ensure conformity of the SSP with the guidelines and procedures for the classification and management of shellfish growing areas as outlined in the NSSP Model Ordinance (2023). Conformity with the Model Ordinance ensures national uniformity of state programs, and the health and safety of shellfish products in intra- and interstate commerce. FDA conducts annual reviews and evaluations of the SSP, and provides recommendations, technical assistance, and training designed to improve the effectiveness of those programs.

C. CALIFORNIA DEPARTMENT OF FISH AND WILDLIFE

The California Department of Fish and Wildlife (CDFW) participates in the SSP through the terms of the “Memorandum of Understanding” (MOU) between the CDFW and the Department of Public Health regarding the California Shellfish and Paralytic Shellfish Poisoning Prevention Program. Wildlife officers within CDFW’s Law Enforcement Division provide assistance to CDPH/PSU in the area of patrol and prevention of illegal shellfish harvesting by both sport and commercial harvesters. In addition, CDFW’s Marine Region and the State Aquaculture Coordinator’s office issues and administers State aquaculture leases for the California Fish and Game Commission.

D. CALIFORNIA CENTRAL COAST REGIONAL WATER QUALITY CONTROL BOARD

The CCRWQCB is responsible for regulating dischargers of wastes to any body of water in the region. The CCRWQCB sets standards for waste dischargers, issues permits to them in the form of Waste Discharge Requirements (WDR) and Monitoring and Reporting Programs (MRP), and monitors their performance.

E. SANTA BARBARA COUNTY ENVIRONMENTAL HEALTH SERVICES

The County Health Officer is responsible for all matters pertaining to public health within Santa Barbara County, including the county’s incorporated cities. The Director of Santa Barbara County Environmental Health Services (SBCEHS) is responsible for enforcing the laws and regulations pertaining to environmental health, including the proper handling and disposal of sewage from private premises.

F. GOLETA SANITARY DISTRICT

The Goleta Sanitary District operates a wastewater collection, treatment, and disposal system that provides service to Goleta Sanitary District, Goleta West Sanitary District, University of California, Santa Barbara, Santa Barbara Municipal Airport, and facilities of Santa Barbara County. The City of Goleta is located approximately eight miles to the west of Santa Barbara. The end of the outfall from the WWTP is approximately 4.1 miles from the aquaculture area.

G. CITY OF SANTA BARBARA, EL ESTERO WASTEWATER TREATMENT PLANT

The City of Santa Barbara, El Estero Wastewater Treatment Plant is a wastewater collection, treatment, and disposal system that provides service to the City of Santa Barbara and portions of Santa Barbara County. The end of the outfall from the WWTP is approximately 4.7 miles from the aquaculture area.

H. SANTA BARBARA MARICULTURE COMPANY

Mr. Bernard Friedman, Santa Barbara Mariculture Company, is the commercial shellfish harvester for Lease M-653-02 Aquaculture Area and was issued a Shellfish Growing Area Certificate by CDPH for commercial shellfish growing and harvesting of oysters, clams, mussels, and scallops at this location. SBMC is listed as a shellstock shipper (#614) on the FDA Interstate Certified Shellfish Shippers List (ICSSL).

V. GROWING AREA CLASSIFICATION

A. CONDITIONALLY APPROVED AREA

The Santa Barbara Lease M-653-02 Aquaculture Area is classified as *Conditionally Approved* for commercial shellfish production in accordance with the NSSP Model Ordinance, Chapter IV (2023). A *Conditionally Approved* area is one that meets the NSSP water quality standards for an *Approved* area (an area from which shellfish may be harvested for direct marketing for human consumption), except during relatively short periods of time when the area does not meet the standards and must be closed to harvesting. Direct marketing means the sale of shellfish harvested without having to undergo purification (relaying or depuration). The factors determining closed periods must be known, predictable, and not excessively complex. The purpose of the *Conditionally Approved* classification is to provide a mechanism, through this Management Plan, for the declaration of harvest closures when pollution impacts to the growing area would cause water and shellfish quality to potentially exceed the standards for an *Approved* growing area.

Conditionally Approved classified areas are eligible as natural wet-storage areas subject to CDPH/FDB wet storage permit requirements. Conditionally Approved areas are eligible as destination locations for relay subject to CDPH/EMB approval and requirements.

B. CLASSIFICATION MAINTENANCE

To maintain the *Conditionally Approved* classification, all Certified Commercial Shellfish Growers shall take a minimum of one water sample per month when in an open status at each primary sample site (Figure 1) using the systematic random sampling (SRS) protocol as directed in the Sampling Plan for Santa Barbara Mariculture Company. In order to obtain rainfall impact data, additional samples should be collected whenever possible during periods of adverse pollution conditions. (See Section XII.A for additional information on water quality sampling requirements).

VI. COMPANY CERTIFICATION

A. REQUIREMENTS FOR CERTIFICATION

Certification of a shellfish growing area is subject to the requirements of Title 17, California Code of Regulations, Sections 7706 to 7733 and Section 112150 et seq. of the California Health and Safety Code. It is unlawful in California to sell, offer, or hold for sale for human consumption any bivalve shellfish unless the harvest area is properly classified and certified by CDPH. A Shellfish Growing Area Certificate is issued to the commercial shellfish company when an application has been accepted for filing, a complete shoreline survey and sanitary survey have been conducted and support this activity, and the growing area in question has been properly classified as *Approved* or *Conditionally Approved*. An area classified as *Restricted* or *Conditionally Restricted* must have an associated relay plan or depuration plant approved by CDPH before a certificate will be issued. The Shellfish Growing Area Certificate expires annually on February 15 and must be renewed each year by submitting an application by January 1.

B. CERTIFICATE COMPLIANCE

The Shellfish Growing Area Certificate contains conditions that the company must meet, including compliance with this Management Plan. If the company holding a current Shellfish Growing Area Certificate violates one or more conditions of the certificate then CDPH may suspend, revoke, or deny renewal of the certificate.

CDPH will conduct field audits and record reviews as part of the ongoing evaluation of company compliance. In accepting a Shellfish Growing Area Certificate the company agrees to cooperate with all requests by CDPH for these audit activities. Companies will be audited on at least an annual basis to ensure that they are performing all responsibilities adequately. Annual audits will include, but not be limited to, evaluation of proper procedures for water quality and biotoxin sampling; maintaining sanitary conditions for all harvest equipment and vessels; maintaining accurate and complete records for all harvest closures, harvest activities and shellstock handling during harvest closures under an approved Standard Operating Procedure (SOP); compliance with the SRS schedule; and compliance with harvest closures. During the annual field audits CDPH can provide orientation and training to new company employees for water quality sampling, shellfish sample preparation for biotoxin analyses, or proper use of field test kits for marine toxins.

VII. PERFORMANCE STANDARDS

A. POTENTIAL NON-POINT SOURCES OF POLLUTION

Non-point sources of pollution are those sources that cannot be attributed to a specific discharge location. Rainfall related sampling established a threshold of three (3) inches of rainfall within any 24-hour period. The data related to the rainfall threshold is presented in the Sanitary Survey. CDPH/PSU also closely monitors the County's beach

monitoring data to determine if the growing area should be closed to harvesting of shellfish. A beach closure in the vicinity of the lease area could cause the closure and/or additional sampling of the growing area.

B. POTENTIAL POINT SOURCES OF POLLUTION

Sources of pollution that can be attributed to a specific site or location are known as “point sources”. Wastewater Treatment Plants fall into this category. The CCRWQCB issues WDRs to all major discharge facilities in the Santa Barbara area. The CCRWQCB publishes copies of the WDRs and annual reports for these dischargers online. These documents are reviewed by CDPH/PSU staff and maintained in the program’s files. There are two WWTPs that discharge treated municipal wastewater to the Pacific Ocean in the Santa Barbara Channel that have the potential to adversely impact the shellfish growing water quality of Lease M-653-02 (Figure 1). Section IV of the Management Plan lists the two treatment plants and provides the location of each treatment plant’s outfall with respect to the growing area. These facilities have WDRs that have effluent limitations on constituents and properties for protection of marine life and human health (Tables 1 and 2). The WDRs for the facilities include a requirement to notify CDPH and any commercial shellfish harvesters in the Santa Barbara Channel in the event of a discharge of inadequately disinfected effluent within 4-hours or by 10 AM the following morning for events after 5 PM and before 6 AM.

C. PROHIBITED AREAS ESTABLISHED

Prohibited areas are those areas where the commercial harvesting of shellfish is not allowed. Water quality in these areas either has not been determined, or does not meet the NSSP standards for shellfish growing waters as determined during the sanitary survey of the area. The size of the *Prohibited* areas for the El Estero ocean outfall and the Santa Barbara Harbor were based on calculations of potential fecal coliform levels and the amount of dilution required to reduce the fecal coliform to acceptable levels. The *Prohibited* areas established around the outfalls of the remaining WWTPs are the calculated distance of 2 miles that a plume of inadequately disinfected effluent could travel in the required notification time of 4-hours assuming a current of 0.82 feet per second in any given direction away from the outfall. This method was more conservative than the results of dilution modeling for these plants.

A *Prohibited* area with a radius of two miles is established around the ocean outfall of the Montecito, Carpinteria, Summerland, and Goleta wastewater treatment plants. A *Prohibited* area with a radius of three miles is established around the ocean outfall of the City of Santa Barbara's El Estero Wastewater Treatment Plant. A *Prohibited* area is established inside the City of Santa Barbara harbor and within a one-half mile radius around the mouth of the harbor.

1. Unclassified Areas

All parts of the Santa Barbara coast area not specifically designated as a certified aquaculture lease are unclassified and therefore considered *Prohibited* areas.

2. Violation of State Laws and Regulations

It is a violation of State laws and regulations to culture, harvest, or hold shellfish intended for sale for human food except in or from areas that have been specifically identified in a valid Shellfish Growing Area Certificate issued by CDPH/PSU. The holding of shellfish in any other waters prior to on-shore processing, storage, and marketing is a prohibited activity. Any shellfish held in uncertified waters, or in water-filled tanks or containers that have not been specifically approved by CDPH/PSU for that purpose, are subject to embargo and/or destruction by CDPH/PSU.

VIII. HARVEST CLOSURE AND NOTIFICATION PROCEDURES

This section provides the closure criteria and notification procedures to be followed by the certified commercial shellfish harvester, and by appropriate State and local agencies, to ensure prompt notification and subsequent closure of the Santa Barbara Lease M-653-02 Aquaculture Area by CDPH/PSU to shellfish harvesting. These procedures shall be followed whenever any pollution event or condition occurs as described below, that is likely to degrade the water quality in the *Conditionally Approved* shellfish growing area to the degree that it does not meet the NSSP minimum standards for an *Approved* area.

A. RAINFALL CLOSURES

Since rainfall closures are an important management tool for the growing area, the harvester should call the Growers' Information line (GIL) prior to harvesting during any period of rainfall. If the harvester is still unsure as to the status of the growing area, they should contact the rainfall duty officer (RDO) at shellfishpreharvest@ccph.ca.gov to determine if the GIL is in the process of being updated (on weekends and off-hours, leave a message on the GIL). CDPH/PSU staff will make all reasonable efforts to ensure that the GIL is current and accurate. However, the GIL is not updated on a continuous basis, and changes that occur in the growing area status due to overnight rainfall will not be updated on the GIL until business hours the following day. If there has been rainfall in the past 24 hours, the harvester shall not harvest unless there is an updated message on the GIL, CDPH/PSU staff states that the growing area is open, or the harvester is able to obtain reliable information that the growing area is open. If the harvester harvests before obtaining current information on the growing area status, and the growing area is closed, the harvester shall be responsible to return all harvested shellfish to the same location it was harvested from.

1. Closure Rules

Rainfall closure rules for M-653-02 shellfish growing area are summarized in Table 3. The M-653-02 shellfish growing area will be closed immediately when the 24-hour cumulative rainfall of 3.0 inches is exceeded as indicated at the National Weather Service rain gauge located at the Santa Barbara Airport (www.crh.noaa.gov/data/obhistory/KSBA.html) or an alternate rain gauge selected by

CDPH/PSU if this one is unavailable. CDPH/PSU will notify the commercial shellfish harvester whenever the rainfall threshold is exceeded and the growing area is closed. The commercial shellfish harvester shall acknowledge receipt of this notification within 24-hours.

The growing area shall be closed for a minimum of three days (72 hours) for rainfall events of greater than 3.0 inches in 24 hours and no greater than 5.15 inches in 24 hours. When determined, an estimated reopening time will be provided. The status of the growing area can be obtained by calling the 24-hour GIL at (510) 412-4644 and/or check the email address provided to CDPH for notifications. If the commercial shellfish harvester harvests before obtaining current information on the growing area status, and the growing area are closed, the commercial shellfish harvester shall be responsible to return all harvested shellfish to the same location it was harvested from. See Section X for the requirements for reopening.

2. Extreme Environmental Conditions

In the case of unusually heavy or prolonged rainfall, exceeding 5.15 inches in a 24-hour period, the closure rules set forth above no longer apply. At the direction of CDPH/PSU, the harvester will collect one water quality sample at the primary station. The earliest possible sample time is no earlier than 24 hours after the end of storm. CDPH/PSU recommends that companies collect the samples for reopening no earlier than 24 hours prior to the posted earliest possible reopening time for the extreme environmental condition event. Leases will be reopened to harvest by CDPH/PSU after acceptable sample results are received, and reopening will occur no earlier than the earliest possible reopening time of the appropriate rainfall closure interval. Fecal coliform results must be equal to or lower than 14 MPN/100 mL for the growing area to be considered for reopening. If results for any sampling station are above the acceptable level, the harvester will be instructed to resample.

If there are subsequent rainfall closure extensions without additional extreme environmental conditions, the sampling requirement remains in place and the earliest possible sample time does not change. Harvesters may satisfy the sample requirement at their discretion but should be aware water quality impacts due to additional rainfall may cause results to be above the acceptable level. The sampling requirement may be removed by CDPH/PSU 21 days after the extreme environmental event end of storm.

3. Movement of Shellstock During Rainfall Closures

The harvest and removal of shellstock from the growing waters during a closure is prohibited. Rainfall closures are necessary for shellstock to depurate to acceptable levels after rainfall events. With an approved Standard Operating Procedure (SOP) there are conditions when shellfish may be moved during a rainfall closure solely for the purposes of sorting and culling prior to being returned to the growing area. Shellstock removed from growing waters during closure periods shall have the number of hours and minutes that the shellstock is out of the water added onto its respective closure time before it can be harvested for market. This shellstock may not be placed in wet storage

and cannot be transported away from the bay or shoreline. This SOP does not apply to seed.

The movement SOP provisions ensure that NSSP patrol requirements can be met to prevent illegal harvesting. Failure to meet the shellstock movement requirements set forth in this Management Plan is a violation of the conditions of the Shellfish Growing Area Certificate issued by CDPH and could result in suspension or revocation of that certificate. Shellstock may not be removed from the water during any other type of closure (e.g. biotoxin closure, sanitary sewage overflow, illness outbreak, etc.). The company shall consult with CDPH to discuss the possibility of shellstock movement during a non-rainfall closure.

There are three identified conditions in which shellstock may be temporarily removed from the growing waters during a rainfall closure. The following paragraphs describe the circumstances of each type of harvest during closure that may be allowed with prior approval:

Condition 1:

Shellstock is removed from the growing water and worked on on-site and then placed back in the growing area. It is NOT removed from the specific commercial shellfish lease/parcel.

Condition 2:

Shellstock is removed from the commercial shellfish harvester's lease/parcel during a closure, transported off the lease/parcel to be worked on, and then returned to the original lease/parcel. This shellstock cannot be transported away from the shoreline.

Condition 3:

Shellstock is harvested during a closure from one *Conditionally Approved* commercial shellfish harvester's lease/parcel and placed into another *Conditionally Approved* lease/parcel, owned by the same commercial shellfish harvester. Both parcels must be under the control and authority of the same commercial shellfish harvester (i.e., product cannot be moved onto another commercial shellfish harvester's lease). *The cleansing time on the destination lease is defined by the longest rainfall closure period of either the source or destination lease.* The shellstock can be either worked on-site on one of the parcels or transported off the lease/parcel to be worked on before being placed back in the growing water. This shellstock cannot be transported away from the bay or shoreline.

NOTE: In all cases shellstock shall not be placed in, or come in contact with, uncertified growing waters.

Standard Operating Procedures (SOPs) Required:

A harvester who intends to move product during rainfall closures under any of the three conditions described above must have a Standard Operating Procedure (SOP) that has

been approved by CDPH. CDPH can provide a template SOP. The approved SOP serves as CDPH approval of the movement *procedures* proposed by the harvester. Removing shellstock from growing waters during a rainfall closure without an approved SOP is a violation of the Shellfish Growing Area Certificate and may result in the embargo of the shellstock, suspension or revocation of the certificate, and legal actions may be taken against the company as allowed by law.

The SOP shall describe how shellstock will be identified and tracked to ensure that product has had the required time in the water for proper cleansing. Shellstock moved during a rainfall closure must be labeled, or the receiving area labeled and demarcated, such that the shellstock can be distinguished to prevent comingling with other product. The SOP shall have an associated tracking form to document the shellstock movements made during a closure. CDPH will approve SOPs that contain acceptable provisions for identifying, isolating, and tracking product that has been moved during rainfall closures.

SOPs must contain the following information:

- a. What type of movement(s) is/are planned; Condition 1, 2 and/or 3; and a description of all movements that are planned.
- b. The leases or growing areas from which shellstock will be removed.
- c. The exact landing and sorting locations wherein shellstock may be sorted or culled.
- d. The leases or growing areas that shellstock will be returned to.
- e. Description of labeling procedures for each bag of shellstock moved, or labeling/demarcating procedures for destination areas that will receive moved shellstock.
- f. Explanation of record keeping, how company will track the movement of product. For shellfish harvested within 1 week of rainfall reopening, a description of how time out of water will be added to the reopening time of moved shellfish and harvest recorded. Include an example of a tracking form to be used to document each shellstock movement on site.
- g. Names of persons responsible for overseeing and documenting shellstock movements.

Tracking forms associated with the SOP must list the following information for each movement:

- a. Date and time shellstock was removed from the growing waters.
- b. Date and time shellstock was returned to growing waters.
- c. Species of shellstock moved.
- d. Type of movement made.
- e. Shellstock source location.
- f. Shellfish landing location - where it was worked on.
- g. Shellstock destination location.

- h. Amount of shellstock moved.
- i. Recorder name or initials.
- j. If harvest within 1 week of reopening:
 - a. Time added to reopening.
 - b. Final harvest date/time and lot number.

B. EMERGENCY CLOSURES: SEWAGE COLLECTION SYSTEM UPSETS

Accidental sewage releases from wastewater collection systems servicing the Santa Barbara area may adversely affect the water quality in shellfish growing areas. The potential for these sewage “upsets” to adversely impact shellfish growing waters is related to the estimated volume of the discharge, the location of the discharge with respect to the growing areas, the prevailing ocean current, and the timing of the discharge with respect to rainfall closures of the growing areas. Criteria for determining whether an emergency closure will be implemented as a result of a sewage spill and criteria for reopening growing areas are presented in Appendix C.

C. EMERGENCY CLOSURES: WWTP MALFUNCTIONS

The *Conditionally Approved* classification of this area is also dependent upon wastewater treatment plant performance. Harvesting of shellfish for human consumption would be determined in part by the performance of the wastewater treatment plants. Discharges of effluent that do not meet current standards shall cause closure of the commercial harvesting area until the impacts can be evaluated. In the event of a sewage spill, or malfunction of a wastewater treatment facility’s disinfection process which results in a potential or actual discharge of inadequately treated effluent into the Santa Barbara Channel, the responsible agency will notify the CCRWQCB, the Santa Barbara Mariculture Company, and CDPH/PSU. CDPH/PSU will then determine the status of the growing area and will notify the harvester when the determination has been made. Harvesting shall not occur until after CDPH/PSU has evaluated the event and determined the required actions. Criteria for determining whether an emergency closure will be implemented as a result of a sewage spill and criteria for reopening growing areas are presented in Appendix C.

D. EMERGENCY CLOSURES: HAZARDOUS SUBSTANCE SPILLS

In the event of a reported hazardous substance spill that could impact a shellfish growing area, the California Emergency Management Agency (Cal-OES) is notified, who in turn will contact the CDPH Duty Officer (on call 24 hours a day). The CDPH Duty Officer will notify the CDPH/PSU supervisor or the next manager in the chain of command. Additional involved agencies will have their own reporting requirements.

Upon notification of a hazardous substance spill CDPH/PSU will determine the status of the growing area and will notify the harvester if a closure is warranted. A harvest closure resulting from a hazardous substance spill shall remain in effect while CDPH/PSU

evaluates the event and determines the required actions. See Section X for reopening requirements.

E. OTHER EMERGENCY CLOSURES

Other emergency situations in which the discharge of a hazardous or deleterious substance to shellfish growing waters is presumed to be likely, but during which communications with CDPH/PSU may be compromised, such as after a natural or man-made disaster, will be evaluated by CDPH/PSU for possible closure of the growing areas.

F. NOTIFICATION PROCEDURES

CDPH/PSU will notify the harvester whenever the rainfall threshold is exceeded and the growing area is closed. The harvester shall acknowledge receipt of this notification within 24 hours. When determined, an estimated reopening time will be provided. The status of the growing area can be obtained by calling the 24-hour GIL at (510) 412-4644. Reopening requirements for rainfall closures are presented in Section X and for sewage spill-related closures in Appendix C. The following notification procedures apply:

1. California Department of Public Health

a. CDPH is the agency responsible for making decisions regarding harvest closures and reopening of shellfish growing areas.

b. CDPH/PSU will maintain a telephone message recorder (the GIL) at the Richmond office, (510) 412-4644, to provide information about closures to commercial shellfish harvesters during non-business hours. CDPH/PSU will update the recorded message as needed and will monitor the line at least twice each day for recorded messages from the harvesters.

c. If CDPH/PSU closes a shellfish growing area to commercial shellfish harvesting, CDPH/PSU will notify each affected certified harvester by telephone or email of that decision. When CDPH/PSU reopens a shellfish growing area to commercial shellfish harvesting, CDPH/PSU will notify each affected certified harvester by telephone or email of that decision. In the case of rainfall closures there are times when storm activities can interfere with the ability of CDPH/PSU to obtain current rainfall information. Beyond normal business hours and weekends CDPH/PSU may not be able to update the closure or reopening information immediately. If a harvester has any question or uncertainty regarding a closure or a reopening, they shall contact the GIL ((510) 412-4644) for guidance prior to resuming any harvest activities. If additional rainfall has occurred since the last GIL update, or if the harvester is unsure about the harvest status of the growing area, they shall consider the area closed to harvest and contact the RDO at shellfishpreharvest@cdph.ca.gov for more information. The harvester should also contact their shellfish growing area specialist during normal business hours for more information. If the harvester harvests before obtaining current

information on the growing area status, and the growing area is closed, the harvester shall be responsible to return all harvested shellfish to the same location it was harvested from.

d. CDPH/PSU will maintain a current contact listing of CDPH/PSU personnel and certified commercial harvesters and provide this listing to the WWTPs, the CCRWQCB, and other applicable parties.

e. Procedures for rainfall closures.

(1) CDPH/PSU will establish a duty roster of staff specialists designated to monitor rainfall at *Conditionally Approved* growing areas, including Santa Barbara. A staff specialist will be designated to be on call for duty as an RDO for every day of the year, including weekends and holidays.

(2) The RDO will monitor weather forecasts and reports to determine whether significant rainfall is predicted or has occurred at the *Conditionally Approved* growing areas. If so, the RDO will contact designated rain gauge stations as needed to obtain current rainfall measurements to determine when the conditions requiring a rainfall closure are met. The designated rain gauge station for rainfall closures for Santa Barbara's conditionally approved growing areas is located at the Santa Barbara Airport in Goleta and can be accessed at www.crh.noaa.gov/data/obhistory/KSBA.html

(3) When the RDO determines that conditions have occurred that require a shellfish harvest closure, he/she will notify each commercial harvester immediately by telephone and email that the area is closed to harvesting and will inform them, if known, when it will reopen.

(4) The RDO will maintain records of rainfall measurements and harvest closure notifications.

f. Procedures for emergency closures:

(1) Upon notification of any pollution event that might be cause for an emergency closure of the Santa Barbara growing area, CDPH/PSU will consult with the responsible agency to consider the extent and severity of the pollution. If an emergency closure is needed to protect sport shellfish harvesters, CDPH/PSU will consult with CDFW and the SBCEHS to determine the location and scope of sport shellfish closures. Criteria for determining whether an emergency closure will be implemented as a result of a sewage spill and criteria for reopening growing areas are described in Appendix C.

(2) CDPH/PSU will notify the harvesters by telephone and email when it declares an emergency closure. When required under H & S § 112160, written confirmation will be sent to the commercial harvesters and copies will be sent to FDB, CDFW, CCRWQCB, the FDA Regional Shellfish Specialist, and any other interested persons. Emergency closure letters shall contain the information required in H & S § 112160(d), so as to afford the recipients reasonable opportunity to submit data, views, or arguments.

(3) As soon as reasonably possible after declaring an emergency closure, CDPH/PSU will initiate an investigation of the effects of the pollution event on water and shellfish meat quality, including such factors as the volume, nature, and location of the discharge, hydrographic factors, and sampling data.

(4) CDPH/PSU will reopen the Lease M-653-02 growing area as soon as it determines that shellfish harvested from the area do not exceed NSSP standards or otherwise pose a public health risk.

(5) CDPH/PSU will notify the commercial harvesters when it decides to reopen the growing area after a closure. When required under H & S § 112160, written confirmation will be mailed to the harvester within five (5) working days after the telephone notification, and copies will be sent to FDB, CDFW, RWCQB, the FDA Regional Shellfish Specialist, and other interested parties.

(6) CDPH/PSU will provide to WWTPs, the CCRWQCB, and to the SBCEHS, a contact list of the certified shellfish harvesters and CDPH/PSU personnel to be called as required in this document. This listing will be updated as needed.

2. Central Coast Regional Water Quality Control Board

The CCRWQCB or its designee will notify CDPH/PSU as soon as possible of any pollution event that may adversely affect water and shellfish quality in the Santa Barbara Channel. Examples of such events are the following: a sewage spill; collection system bypass; malfunction of a WWTP, WWRF or any discharge of a hazardous chemical, pesticide, or petroleum product. The CCRWQCB will include the following mandatory notifications in its draft Waste Discharge Requirements issued to the WWTPs listed in Table 1.

The permitted discharger will notify CDPH/PSU, and SBMC immediately; and SBCEHS as soon as possible of any sewage spill, collection system bypass, or malfunction of a WWTP or WWRF which results in a potential or actual discharge of raw or incompletely treated sewage which may adversely affect water and shellfish quality in Santa Barbara Channel; and;

The discharger will develop and maintain written procedures incorporating the notification requirement described above, to post the procedures at the facility, and to provide a copy of the current notification procedures to CCRWQCB, CDPH/PSU, and SBCEHS.

3. Santa Barbara County Environmental Health Services

SBCEHS will notify CDPH/PSU as soon as possible of any discharge of sewage into the Santa Barbara Channel, directly or indirectly, when they become aware that such an event has occurred. SBCEHS monitors the water quality at public beaches for bacterial concentrations as required by H & S § 115880 (AB 411) on a weekly basis and forwards

the results to CDPH/PSU. If the water adjacent to a public beach fails to meet established bacterial standards, SBCEHS will post signs alerting the public that the water does not meet health standards. The SBCEHS weekly report includes information on beach postings and closures.

4. City of Santa Barbara's El Estero Wastewater Treatment Plant and Goleta Sanitary District

a. In the event of a sewage spill, or malfunction of a wastewater treatment facility's disinfection process which results in a potential or actual discharge of inadequately treated effluent into the Santa Barbara Channel, the responsible agency will notify CCRWQCB, the Santa Barbara Mariculture Company, and CDPH/PSU as soon as possible after becoming aware of the event. All monitoring data related to the event shall be provided to CDPH/PSU. Note: the specific notification procedure has not been formalized at this time with Goleta Sanitary District. See paragraph IX.F.4c below for the interim procedure.

b. The agencies named above will develop and maintain written procedures incorporating the notification requirement, post the procedures at their facilities, and provide a copy of the current procedures document to CCRWQCB and CDPH/PSU.

c. The MOU between the City of Santa Barbara and CDPH establishes the notification procedure between the El Estero WWTP and the commercial shellfish harvesters in the area. CDPH does not have a similar agreement with Goleta. The CCRWQCB has agreed to include a notification procedure in each WDR/MRP. CDPH/PSU has requested the General Managers at the four WWTPs to notify CDPH/PSU and each certified commercial shellfish harvester in the region as soon as possible of any effluent discharge or any malfunction of the wastewater treatment facility's disinfection process which results in a potential or actual discharge of inadequately treated effluent into the Santa Barbara Channel.

5. Impact Assessment

The responsible party shall provide sufficient information to CDPH/PSU such that the magnitude of the impact to the growing area can be assessed. This information should include, but not be limited to, the location of the discharge, a detailed characterization of the discharge when petrochemicals or other hazardous materials are involved, the rate and volume of the discharge, the point of discharge into the Santa Barbara Channel, and the identification and flow rate of any receiving waters, when appropriate. It is recommended that water samples be collected for the appropriate analysis from the area of impact, as well as from nearby sites outside the zone of impact. Under some circumstances it may also be necessary to analyze shellfish tissue samples to assess the impact to the product in the growing areas.

IX. CERTIFIED SHELLFISH HARVESTER RESPONSIBILITIES

This section contains the major responsibilities of the certified shellfish harvester; additional responsibilities are included elsewhere in the Management Plan.

A. RECORD KEEPING

The Santa Barbara Mariculture Company shall keep records showing the dates and times of all harvest closures, re-openings, and harvest operations. These records shall be kept on a form similar to that shown in Appendix D. If the certified commercial shellfish harvester has an approved rainfall movement SOP, records for movement during a rainfall closure shall be maintained. Completed forms shall be maintained in a file at the harvester's place of business for no less than two years. One copy of the completed form shall be submitted to the CDPH/PSU upon request. Each commercial shellfish harvester should be prepared to present each monthly log and SOP forms to the CDPH, local health agency, or CDFW during any inspection.

The Santa Barbara Mariculture Company shall report harvest quantities by species to CDFW. The reporting shall be at a frequency not to exceed monthly.

B. HARVEST CLOSURE

The Santa Barbara Mariculture Company shall cease harvesting operations when notified by CDPH/PSU of a harvest closure or when otherwise required by the closure rules and procedures set forth in this Management Plan. Compliance with this Management Plan is mandatory as a condition of the Shellfish Growing Area Certificate issued by CDPH/PSU. Any violation of a harvest closure declared pursuant to this Management Plan may be cause for suspension or revocation of a certificate or for other disciplinary actions as provided by law. If the harvester has any questions or uncertainties regarding a rainfall closure, they should contact CDPH/PSU for guidance or clarification prior to commencing harvesting or marketing operations. Harvester procedures for closures associated with sewage spills are described in greater detail in Appendix C.

1. Business Hours

During normal State business hours (Monday-Friday 8AM-5PM, excluding State holidays), the harvester should call the CDPH/PSU field staff at (510) 412-4638. If a shellfish specialist cannot be reached the harvester should try alternate contacts on the most current CDPH staff call down list (Appendix A).

2. Non-Business Hours

During non-business hours, the harvester should call the CDPH/PSU commercial shellfish GIL. After listening to the recorded message, if more information is needed, the harvester should leave a message or send an email to the program at shellfishpreharvest@cdph.ca.gov.

C. RECEIVING AND ACKNOWLEDGING NOTIFICATIONS

The Santa Barbara Mariculture Company shall maintain a 24-hour telephone line with a recorder or voice mailbox and provide the telephone number to CDPH/PSU. If notified of a closure by CDPH/PSU via the telephone message, the harvester shall acknowledge receipt of that message within 24 hours by email to shellfishpreharvest@cdph.ca.gov, return call to the CDPH/PSU shellfish specialist at (510) 750-2554 or the CDPH/PSU shellfish GIL at (510) 412-4644. If a recorder answers these lines, after the recorded message the commercial harvester shall leave a message acknowledging receipt of the closure notification.

D. OYSTER HARVEST IN RELATION TO VIBRIO PARAHAEMOLYTICUS

Each certified commercial shellfish harvester is responsible for enacting control measures for oyster harvest to reduce risk of *Vibrio parahaemolyticus* (Vp) illness. The harvester shall comply with the terms, conditions, and procedures outlined in the 2021 *Vibrio parahaemolyticus* Control Plan for Oysters in Commercial Shellfish Growing Areas of California (VpCP) as well as submit monitoring samples requested by CDPH.

The VpCP establishes maximum allowable times from harvest to temperature control (time to temperature control) for each growing area by month. Certified commercial shellfish harvesters shall place harvested oyster shellstock into temperature control (e.g. cold storage) or into depuration/wet storage tanks within the maximum allowable times. For the purposes of time to temperature control, the start of harvest is the time of exposure either by tidal action or by removal from water and placement onto vessel. Intertidal growing areas need to record tidal exposure time as the harvest time in harvest records as applicable.

E. REPORTS OF POLLUTION OR ILLNESS

1. Illness

If the harvester has knowledge of any suspected illness resulting from the consumption of shellfish harvested from the growing area, the commercial harvester shall notify CDPH/PSU and CDPH/FDB immediately.

2. Sewage or Hazardous Substance Spill

If the Santa Barbara Mariculture Company has knowledge of any potential sewage or hazardous substance spill which may adversely affect water and shellfish quality in the Santa Barbara lease area, the commercial harvester shall close the growing area to harvesting and shall notify CDPH/PSU immediately by telephone.

a. Business Hours

During normal business hours the commercial harvester shall contact CDPH/PSU staff using the call-down list provided in Appendix A.

b. Non-Business Hours

During non-business hours, or if CDPH/PSU staff cannot be reached, the harvester shall call the California Emergency Management Agency in Sacramento at (800) 852-7550 or (916) 262-1621. Provide your name, firm, and telephone number, the reason for the call, and ask to be contacted by a staff person from the CDPH/PSU. The harvester should also email the shellfish program at shellfishpreharvest@cdph.ca.gov with details of the reported spill.

3. Wastewater Treatment Plant Notification

In the event of a malfunction of a wastewater treatment facility's disinfection process that results in a potential or actual discharge of inadequately treated effluent into the Santa Barbara Channel, the responsible agency will notify the harvester as soon as possible. CDPH/PSU will determine the status of the growing area and will notify the harvester when the determination has been made. Harvesting shall not occur until after CDPH/PSU has evaluated the event and determined the required actions. If the harvester has been notified by a WWTP, the CCRWQCB, SBCEHS or any other reliable source of a sewage or hazardous substance spill which may adversely affect water and shellfish quality in the Santa Barbara Channel, and if the harvester cannot contact CDPH/PSU immediately, they shall consider the growing area closed and shall suspend harvesting operations immediately, and shall not resume such operations until approval to do so has been received from CDPH/PSU.

4. Liveboards

SBMC shall immediately report the presence of any moored boat that is located within the safe distance allowed in Figure 2 to the growing area to the CDPH/PSU shellfish specialist.

F. VESSEL SANITATION AND HUMAN WASTE

1. Vessel Sanitation

Vessels used to harvest and transport shellstock must be properly constructed, operated, and maintained to prevent contamination, deterioration, and decomposition of the shellstock. Decks and storage bins shall be constructed and located to prevent bilge water or unclassified/Prohibited area overboard water from coming into contact with the shellstock. Bilge pump discharges shall be located so that the discharge shall not contaminate shellstock.

Boat decks and storage bins used in the harvest or transport of shellstock for direct marketing shall be kept clean with potable water or water from a growing area in the

open status of the Conditionally Approved classification and provided with effective drainage.

Vessels and all other equipment coming in contact with shellstock during handling or transport for relaying or depuration shall be thoroughly cleaned before the vessels or equipment are used to transport or handle shellfish for direct marketing.

When necessary, effective coverings shall be provided on vessels to protect shellstock from exposure to hot sun, birds, and other adverse conditions. Cats, dogs, and other animals shall not be allowed on vessels.

2. Disposal of Human Sewage from Vessels

Human sewage shall not be discharged overboard from a vessel used in the harvesting of shellstock. An approved marine sanitation device (MSD), portable toilet, or other sewage disposal receptacle shall be provided on the vessel to contain human sewage.

Portable toilets shall be used only for the purpose intended, be secured while on board and located to prevent contamination of shellstock by spillage or leakage, be emptied only into a sewage disposal system, be cleaned before being returned to the boat, and not be cleaned in equipment used for washing or processing food. Sewage receptacles other than MSDs or portable toilets must be constructed of impervious, cleanable materials, have tight fitting lid, and be indelibly labeled "Human Waste" in contrasting letters at least 3 inches in height

G. AQUACULTURE OPERATIONS

1. Seed

The certified shellfish harvester shall obtain approval from CDPH/PSU of all sources of seed shellstock, in accordance with the NSSP. Records of all seed purchases shall be maintained and provided to CDPH upon request.

Harvesters shall assure that seed obtained from Prohibited or unclassified waters does not exceed the maximum seed size as defined in Section II. Note that shellfish seed may be defined differently in other states or countries. Shellstock exceeding maximum seed size are no longer considered seed and are treated as marketable shellstock.

If the shellfish harvester is cultivating seed in *Prohibited* or unclassified waters, a Seed Operational Plan must be in place and approved by CDPH in accordance with the 2023 NSSP. The Seed Operational Plan outlines requirements for seed cultivation in unclassified waters and procedures for destruction of shellfish that exceeds seed size.

2. Shellstock Movement into Certified Growing Areas

Shellstock over seed size shall not be moved between different water bodies and into certified growing areas except under certain relay or wet storage activities requiring a

permit and approval from CDPH. Shellstock over seed size shall not be moved from *Prohibited* or unclassified waters into certified growing areas.

a. Transplant Operational Plan

Shellstock may be transplanted for grow-out between certified growing areas (in different bodies of water) with a CDPH approved Transplant Operational Plan and under specific conditions. Transplanting shellstock is only allowed from a *Conditionally Approved* or *Approved* growing area in open status to another certified growing area. The purpose of transplanting is to grow-out shellstock at the destination location for a minimum 60 days, long enough that the destination location becomes the harvest location and there are no illness traceability concerns.

The approved Transplant Operational Plan serves as CDPH approval of the procedures proposed by the harvester. CDPH will approve Transplant Operational Plans that contain acceptable provisions for identifying, isolating, and tracking shellstock. A template operational plan can be provided by CDPH. Separately, additional approvals for transferring shellstock may be needed from CDFW.

Transplant conditions:

- 1) The harvester must have a CDPH-approved Transplant Operational Plan.
- 2) The source of shellstock is *Approved* or *Conditionally Approved* in open status.
- 3) The transplanting destination location is a certified growing area.
- 4) The shellstock must remain in the destination location for a minimum of 60 days.
- 5) During the first 60 days after planting at the destination location, the transplanted shellstock:
 - a. cannot be containerized for sale,
 - b. must be held separate from shellstock at the destination location and not co-mingled, and
 - c. must be marked for identification.
- 6) Required documentation will include: source information, location of harvest (source location), dealer #, date/time of harvest, growing area status, specific location planted, date/time planted, attestation of status at 60 days post-planting, and any other applicable tracking information.
- 7) The shellstock for transplant must have identifying documentation during transport.
- 8) Documentation and transplanted shellstock are subject to inspection.
- 9) After 60 days at the destination location, the source of the transplanted shellstock is considered to be the destination location.

Harvester adherence to the procedures set forth in the Transplant Operational Plan will be verified through growing area inspections and audits. Commercial shellfish harvesters shall maintain a field log in the work area for documenting the current shellstock transplant activities. The transplant records shall be provided upon request.

3. Aquaculture That Attracts Birds or Mammals

If CDPH determines that the aquaculture structures attract sufficient birds and/or mammals such that their waste presents a human health risk, the Company shall have a separate Wildlife Operational Plan document (NSSP, 2023).

Fecal contamination from wildlife to a shellfish growing area can be erratic yet significant. If the CDPH/PSU has determined that wildlife are significantly impairing the water quality of shellfish growing waters then the affected commercial shellfish harvester must have on file with CDPH/PSU an approved Wildlife Operational Plan to prevent product contamination detailing methods used to mitigate fecal contamination of the product by wildlife. The plan shall include species harvested, a description of the facilities and harvest operation, the types and locations of any structures in the growing waters, and specific procedures and time frames of deterrents and activities for purposes of mitigating wildlife impact on marketable product.

It is the harvester's responsibility to consult with the appropriate agencies to determine which measures are acceptable and legal in preventing wildlife from contaminating shellfish. All harvesters who are required by the CDPH/PSU to have a Wildlife Operational Plan are responsible for obtaining any state or federal permits that may be required to prevent wildlife from contaminating shellfish. The plan must have approval by CDPH/PSU prior to implementation and shall be revised as procedures change.

H. AUDITS

The harvester shall comply with all field audits and harvest log reviews. The harvester will accommodate CDPH in all aspects of the field audits including but not limited to: scheduling the audit, providing safe boat support to the growing area, and collecting samples at primary sites. Failure to comply with the audits could result in suspension or revocation of the shellfish Growing Area Certificate issued by CDPH/PSU.

I. REPORTS OF ILLEGAL ACTIVITIES

Certified shellfish harvesters are responsible for all activities on their leases, including the legal disposition of all marketable shellstock. If a certified commercial shellfish harvester has knowledge or suspicion of possible illegal harvesting activities, theft of marketable product, trespassing by persons unaffiliated with the company, illegal subleasing, or placement of commercial shellstock in any *Prohibited* area, the commercial harvester shall notify CDPH/PSU and CDPH/FDB immediately.

X. REOPENING PROCEDURES

A. RAINFALL CLOSURES

Prior to reopening following a closure due to rainfall, the harvester shall contact the GIL (510 412-4644) or review status emails to determine if any changes have occurred in the reopening times for each growing area. If there is any uncertainty about the message or if no message is found on the GIL the harvester shall contact the RDO at (510) 412-4638 or email the shellfish program. The harvester is advised that reopening

dates and times are subject to change. CDPH/PSU staff will make all reasonable efforts to ensure that status updates are current and accurate. The GIL is not updated on a continuous basis, and changes that occur in the growing area status due to overnight rainfall will not be updated on the GIL until business hours the following day.

B. SEWAGE OR HAZARDOUS SUBSTANCE SPILLS

1. Investigation

As soon as reasonably possible after declaring an emergency closure, CDPH/PSU will initiate an investigation of the effects of the pollution event on water and shellfish meat quality, including such factors as the volume, nature, and location of the discharge, hydrographic factors, and sampling data. Criteria for determining whether an emergency closure will be implemented as a result of a sewage spill are described in Appendix C.

2. Harvest Reopening

Criteria for reopening growing areas after a sewage spill are described in Appendix C. Due to the variable and unpredictable nature of hazardous material spills, reopening criteria for these events will be determined on a case by case basis.

3. Notification

CDPH/PSU will notify the harvester and CDFW when it decides to reopen the growing area after an emergency closure. When required under H & S § 112160, written confirmation will be mailed to the harvester within five (5) working days after the telephone and email notifications, and copies will be sent to FDB, CDFW, RWCQB, the FDA Regional Shellfish Specialist, and other interested parties.

C. REOPENING WHEN SAMPLING IS REQUIRED

Growing areas may be closed for extended periods of time due to extreme environmental conditions, samples containing elevated fecal coliform concentrations, or sewage spills. Water quality sampling may be required before reopening during these circumstances to demonstrate that water quality has returned to acceptable levels. Growing areas that are closed and require sampling to reopen will be reopened effective the date and time that PSU staff receive and have reviewed acceptable results from the laboratory.

Criteria for reopening growing areas following extreme environmental conditions are presented in Section VIII.A.2 of this Management Plan, criteria for reopening following elevated FC concentrations are presented in Section XII.D and criteria for sewage spill closures are presented in Appendix C.

XI. PREVENTION OF ILLEGAL HARVESTING (PATROL)

A. ILLEGAL HARVESTING OF SHELLFISH

During harvest closures implemented by CDPH, CDPH/PSU staff coordinates regular patrols of shellfish growing areas to prevent illegal harvesting activity on the part of harvesters or unauthorized individuals. In addition to this Management Plan, three other documents describe the activities of CDPH/PSU (including that part of the program administered by FDB) and CDFW in preventing illegal harvesting of shellfish, preventing the movement of contaminated and/or toxic shellfish into avenues of commerce, and for ensuring the proper handling and identification of shellfish after harvest. These documents are: (1) the MOU between CDFW and CDPH, signed June 2016; (2) the CDPH "Patrol Policy for Commercial Shellfishing in California"; and (3) the CDPH "Contingency Plan for Marine Biotoxins in California Shellfish".

B. DEPARTMENT OF FISH AND WILDLIFE, LAW ENFORCEMENT DIVISION

CDFW's Law Enforcement Division field office for the Santa Barbara area is headquartered at 1933 Cliff Drive, Suite 9, Santa Barbara, CA. CDFW maintains patrol of areas with significant naturally occurring shellfish resources for illegal commercial harvesting. CDFW patrol activities may occur at any time. CDFW officers have a broad range of enforcement duties, shellfish being only one part of their responsibilities.

The provisions of this Management Plan as it applies to the obligations of the commercial shellfish harvester are enforced by CDPH/PSU, not CDFW. CDFW agrees to assist CDPH/PSU, when specifically requested by CDPH/PSU, by redirecting some of its patrol effort to monitor commercial harvesters for compliance with harvest closures. If CDFW patrol personnel find or suspect any commercial shellfish harvesting in violation of a harvest closure established pursuant to this plan, CDFW will report their finding to CDPH/PSU.

C. ILLEGAL MARKETING

CDFW also makes periodic checks for the illegal marketing of uncertified shellfish by means of market and restaurant inspections. This activity also is carried out by local environmental health agencies. Shellfish in the marketplace are inspected for proper identification and records of purchase. The SBCEHS or the State CDPH/FDB can confiscate and destroy any shellfish that lack proper identification or have been subject to improper handling or storage. CDFW can take action on shellfish that lack proper identification.

D. HARVESTER COMPLIANCE WITH CLOSURES

(1) CDPH/PSU (or CDFW patrol personnel, or FDB inspectors, upon request from CDPH/PSU) will field check the harvest area to look for harvesting activity during closure periods.

- (2) CDPH/PSU will review harvester records of closure and harvesting dates.
- (3) CDPH/PSU will spot check by telephone to ascertain whether harvester is conducting operations during closed periods.
- (4) CDPH/PSU will maintain records of all compliance monitoring activities and findings.

XII. WATER QUALITY MONITORING

The Santa Barbara Mariculture Company is responsible for providing water quality monitoring data from its growing area, as required by the NSSP Model Ordinance, Chapter IV (2023), with coordination and oversight from the CDPH/PSU. The harvester submits samples according to a sampling plan approved by CDPH/PSU. Failure to maintain the required sampling schedule is a violation of the NSSP requirements and may result in reclassification of the growing area to *Prohibited* as well as disciplinary actions including, but not limited to, the revocation and or suspension of the growing area certificate as provided for by law. The following is a summary of the requirements associated with routine monitoring of shellfish growing waters.

A. WATER SAMPLING REQUIREMENTS

Water samples are collected at the primary site on the northeast corner of the lease (Figure 1) at least bimonthly during periods when the growing area is open for harvesting. Sampling shall be accomplished using the systematic random sampling protocol in accordance with the current Sampling Plan for Water Quality Monitoring. The harvester shall also consider any adverse conditions when determining the proper time of day to collect samples (e.g., ebb tide if this increases the potential of capturing water quality impacts from potential pollution sources). If a sampling date occurs during a rainfall closure, the sampling shall be conducted on the first open day (i.e., the day of reopening). The samples shall be tested at a CDPH/PSU approved laboratory for fecal coliform assay for water and shellfish, and the results sent directly to CDPH/PSU. If a sampling date occurs during a closure, the sampling shall be conducted on the first open day (i.e., the day of reopening). In order to obtain data on rainfall-related impacts to water quality, additional samples should be collected whenever possible during periods of adverse pollution conditions, such as immediately following rainstorms or on the first days of reopening following a closure.

All compliance sampling is to be conducted by a commercial shellfish harvester staff member that has been trained by the CDPH/PSU. If sampling has not been conducted by a CDPH trained staff member then the sampling results will be rejected by the CDPH and the commercial shellfish harvester will be required to resample with trained staff. CDPH will evaluate all certified samplers annually and provide training for compliance sample collection upon request from the commercial shellfish harvester.

The harvester shall notify CDPH/PSU in writing prior to the scheduled sampling date when sampling will not occur on the scheduled date (and the area is in the open status) and shall request an alternative date. A reason for the deviation from the Sampling

Plan schedule must be provided. If CDPH/PSU approves the request, a new date will be assigned. If CDPH/PSU does not approve the request, the harvester will be required to sample on the original date specified in the schedule. Samples collected on incorrect dates will not be identified as a compliance samples, and will not be used in the calculations for verifying classification of the area.

B. TRANSPORTATION

The harvester provides his own boat and all necessary sampling equipment for routine water quality monitoring. Boat trips for CDPH/PSU staff may be scheduled with the harvester for additional survey work, special studies, and to audit water sampling technique by the harvesters. Harvesters shall be responsible for transporting samples to the certified laboratory and meeting proper holding times and temperature limitations as detailed in the sampling plans.

C. LABORATORY

The harvester shall submit water samples to a laboratory certified for testing shellfish growing area waters. A certified laboratory is one that has been evaluated for the approved shellfish growing water method(s) by the FDA or an FDA-approved Laboratory Evaluation Officer and found to be in conformance with the NSSP requirements. Analytical results are to be transmitted directly from the laboratory to CDPH/PSU.

D. WATER SAMPLES EXCEEDING NSSP CRITERIA

Water sample results that exceed NSSP criteria may indicate a public health threat, and must be responded to immediately. A two-tiered response protocol is described in the following paragraphs. *In any case when a sample exceeding NSSP criteria results in a growing area closure, the closure shall be effective at the date and time the sample was collected. Reopening following satisfactory sample results shall be effective at the date and time that sample results are received from the laboratory.*

Any sample collected from an open area that contains fecal coliform concentrations *greater than 43 MPN/100 mL but less than 107 MPN/100 mL*¹ must be re-sampled immediately. Re-sampling must be conducted within 24-hours of notification of initial sample results². If re-sampling is not conducted within the specified time frame, the portion of the lease or growing area represented by the sampling station in question shall be closed for harvest until re-sampling is conducted and results demonstrate that fecal coliform concentrations have returned to acceptable levels. The commercial shellfish harvester should refrain from harvesting from the area represented by the

¹ The 99th percentile of a population with a geometric mean of 14 MPN/100 mL and a 90th percentile of 43 MPN/100 mL

² This requirement is contingent on laboratory support and growing area status. CDPH recognizes that re-sampling cannot be conducted when the lab is not accepting samples (e.g., on weekends and holidays), but must be conducted as soon as possible when the lab reopens. If the growing area closes due to rainfall, re-sampling should be conducted on the day of scheduled reopening.

sampling station until re-sample results are received and are acceptable relative to commercial growing areas standards.

If re-sampling results are greater than 43 MPN/100 mL fecal coliform, the represented portion of the lease or growing area shall immediately be closed to harvest. The area will be considered for reopening when further re-sampling produces a result below 14 MPN/100 mL fecal coliform. In certain cases, CDPH staff may consider reopening based on a re-sample result equal to or below 43 MPN/100 mL *combined with a declining trend in fecal coliform levels at the compliance sampling station and surrounding areas*. In order to establish fecal coliform trends in surrounding areas it is recommended that re-sampling be conducted both at the primary compliance station(s) and other surrounding water quality stations. CDPH will advise the harvester as to which water quality stations should be sampled.

A sample collected from an open area that contains *greater than or equal to 107 MPN/100 mL* fecal coliform shall result in immediate closure of the portion of the growing area represented by the sampling station from which the elevated sample was collected. The area will be considered for reopening when re-sampling produces a result equal to or below 14 MPN/100 mL fecal coliform or a result equal to or below 43 MPN/100 mL *combined with a declining trend in fecal coliform levels at the compliance sampling station and surrounding areas*. Reopening criteria are as described above.

If re-sampling continues to yield elevated fecal coliform levels CDPH/PSU will conduct an investigation, which may include additional sampling, to confirm the elevated indicator levels and determine the source, and will notify all relevant regulatory agencies. If a source is identified, CDPH/PSU will refer the matter to the appropriate regulatory agency for corrective action. The commercial shellfish growing area will remain closed until the pollution problem is identified and remediated, and the commercial shellfish growing area is determined to be free of pathogens and to meet NSSP standards, or until the area is correctly reclassified by CDPH/PSU. CDPH will analyze the most recent water quality data and determine if the current shellfish growing area classification is correct or if the commercial growing area needs to be closed or reclassified.

XIII. MARINE BIOTOXIN MONITORING

A. CONDITIONS AND PROCEDURES

The Santa Barbara Mariculture Company shall comply with the following conditions and procedures for management of marine biotoxins. All commercial shellfish harvesters shall follow the requirements of the “Marine Biotoxin Monitoring and Control Program Field Sampling Protocol: Shellfish for Commercial Shellfish Harvesters” protocol when sampling is required (Appendix E). Biotoxin sampling shall be conducted prior to and during any harvest activities as described below. If a sample cannot be collected the commercial shellfish harvester must notify the CDPH.

Special management protocols with Alert Stages (Tables 3 and 4) detail response activities for different concentrations of each biotoxin. The protocols detail the specific requirements for maximum time between sampling and harvest.

When there is a pattern of positive field tests for domoic acid and when lab resources are readily available, CDPH may permit SBMC to discontinue field testing for this toxin at the company's discretion. Testing shall resume when the event has subsided and there are two consecutive non-detect domoic results or when the harvester is instructed by CDPH.

1. Phytoplankton Observations

In addition to shellfish samples, it is strongly recommended that SBMC collect at least weekly samples of phytoplankton for the early detection of toxic blooms. As shellfish samples are not tested for all possible biotoxins, the phytoplankton samples will provide information about the presence of potentially toxic blooms (i.e., *Pseudonitzschia* and *Dinophysis*, which produces diarrhetic shellfish poisoning toxins) CDPH will provide training in the sampling and field identification of the toxic species and can provide quality control by examining samples and comparing observations with the company.

2. Paralytic Shellfish Poisoning Analysis

SBMC shall submit at least weekly samples of all harvested species to the CDPH Microbial Disease Laboratory, or another laboratory, at the harvester's expense, found to be in conformance with the NSSP requirement for PSP toxin analysis by an FDA Laboratory Evaluation Officer (LEO) or FDA-certified LEO. A sample of each species shall be collected at the current depth of harvest. Samples shall be collected as closely as possible to time of harvest. If the CDPH Microbial Disease Laboratory is available, all samples shall arrive at the laboratory by Tuesday or Wednesday morning but no later than Thursday morning 10:30 AM¹. More frequent sampling may be required by CDPH to adequately track an impending or existing PSP toxin event. The samples taken from the actual growing area will be evaluated relative to the thresholds and time frames located in Table 4.

3. Domoic Acid Analysis

SBMC shall submit at least weekly samples of all harvested species for domoic acid analysis. CDPH/PSU will submit the samples from the current week collected for PSP toxin analysis to the CDPH Food and Drug Laboratory for domoic acid analysis. If the CDPH Food and Drug Laboratory is available, the commercial shellfish harvester should coordinate sample arrival with CDPH. The samples may need to arrive as early as Tuesday morning by 10:30 AM to be processed in time. If the CDPH Food and Drug Laboratory is unavailable, the harvester shall submit samples as directed by CDPH, at the harvester's expense, to an alternate laboratory found to be in conformance with the NSSP requirements for domoic acid analysis by an FDA LEO or FDA-certified LEO. A

¹ Subject to laboratory availability.

sample of each species shall be collected at the current depth of harvest. Samples shall be collected and submitted as closely as possible to the time of harvest. At the discretion of CDPH, more frequent sampling may be required to adequately track an impending or existing domoic acid event. The samples taken from the actual growing area will be evaluated relative to the thresholds and time frames located in Table 3.

The company may use a commercially available diagnostic domoic acid test kit to screen shellfish samples for domoic acid. For the results to be considered by CDPH as a valid screening result, the company must first demonstrate proficiency with the method to CDPH.

4. Diarrhetic Shellfish Poisoning Analysis

There is the potential for diarrhetic shellfish poisoning (DSP) toxins to occur in commercial shellfish. Cells of the dinoflagellate genus *Dinophysis* (the genus includes species that produce DSP toxins) are often observed in phytoplankton samples from California coastal areas, usually at low levels.

The CDPH/PSU does not have the resources to test for Diarrhetic Shellfish Poisoning (DSP) toxins and has established the following threshold of 34,500 cells per liter for *Dinophysis* concentrations in representative phytoplankton samples that when exceeded will trigger the request for testing of shellfish from the growing area for DSP toxins. CDPH/PSU will prescreen phytoplankton samples with the established cell biomass estimate of a relative abundance index¹ (RAI) to determine when total *Dinophysis* cells should be quantified in cells per liter. When the calculated RAI for total *Dinophysis spp.* exceeds 0.056 then CDPH will require a whole water sample be collected for the cell count method. CDPH will conduct the cell counts on whole water samples for total *Dinophysis* to determine if the 34,500 cells per liter threshold is exceeded. CDPH/PSU will provide the necessary equipment, shipping materials, and sampling instructions to SBMC for the collection and submission of a whole water sample.

If counts for *Dinophysis spp.* exceed 34,500 cells per liter in the Santa Barbara Channel CDPH/PSU may require SBMC to submit samples of harvested species for DSP toxin analysis at the harvester's expense. CDPH will assist the harvester to identify an appropriate laboratory for DSP toxin testing. Sampling shall continue until DSP toxins and *Dinophysis* levels fall to acceptable levels as determined by CDPH/PSU. The laboratory results for reopening the growing area under Section XIII. G below must be from a laboratory certified by the FDA for the NSSP approved DSP method if the growing area is closed to harvest due to DSP levels in shellfish exceeding the alert level of 0.16 ppm.

5. Sampling Protocol

¹ The Relative Abundance Index is the percent composition of the sample represented by the genus multiplied by the settled volume of the bulk sample in milliliters divided by the tow length for the sample in feet. The units are mL per foot of seawater towed.

SBMC shall follow the field sampling protocol for marine biotoxins provided by CDPH for sample collection, preparation, and submission (Appendix E). Biotoxin samplers are trained and audited annually by CDPH. Bernard Friedman of SBMC was audited for biotoxin sampling on November 2, 2022.

6. Additional Sampling

SBMC shall increase the sampling frequency, expand sampling to include any other commercial shellfish species designated in the Shellfish Growing Area Certificate, and/or cease harvesting as directed by CDPH.

B. CLOSURE CRITERIA

When a sample exceeds the biotoxin alert level, the growing area shall be closed immediately, effective on the date and time the sample was collected. FDB will be notified and will be responsible for decisions regarding product recall or destruction.

1. Domoic Acid Alert Level

In compliance with the NSSP Model Ordinance, the alert level for domoic acid is reached when the concentration of the toxin in shellfish meat equals or exceeds 20 micrograms per gram of tissue (i.e., 20 parts per million [ppm]).

2. Paralytic Shellfish Poisoning Alert Level

In compliance with the NSSP Model Ordinance, the alert level for PSP toxin is reached when the concentration of the toxin in shellfish meat equals or exceeds 80 micrograms per 100 grams of tissue.

3. Diarrhetic Shellfish Poisoning Alert Level

In compliance with the NSSP Model Ordinance, the alert level for DSP is reached when the concentration of the DSP toxins in shellfish meat equals or exceeds 0.16 milligrams per kilogram (mg/kg) of tissue (i.e., 0.16 ppm or 16 µg/g).

4. Samples Exceeding the Alert Level

A biotoxin concentration at or above the respective alert level in one or more samples of commercial product in the growing area prior to harvest, or in commercial channels after harvest, will trigger an immediate closure of the growing area until follow-up sampling confirms that the toxin has returned to safe levels as described in Section G below, and in Tables 3 and 4. If notified of a biotoxin-related closure or harvest restriction by CDPH/PSU via telephone message recorder, the harvester shall acknowledge receipt of the message within 24-hours by return call to the CDPH/PSU shellfish specialist at (510) 412-4638, the unit supervisor at (510) 412-4635, or the CDPH/PSU shellfish growers information line (GIL) at (510) 412-4644.

5. Failure to Submit Required Samples

It is the responsibility of SBMC to collect and ship the required samples to a certified laboratory for biotoxin analysis. Failure to submit the required samples or inability of the analytical laboratory to provide results in a timely manner may result in an immediate harvest closure. CDPH will notify FDB, who will determine if a product recall is warranted. The reopening criteria for closure due to lapse of sample is in Section G.

6. Screening Tests

If a sample is tested with an NSSP approved screening test for PSP or domoic acid and is positive, and a quantitative result cannot be reported in a timely manner, the growing area may be placed in a precautionary closure. The precautionary closure will be removed in the presence of a quantitative result less than the closure threshold.

C. ALERT STAGE HARVEST RESTRICTIONS

The special management protocols in Tables 3 and 4 outline multiple alert stages for concentrations of biotoxins below the closure alert level. The tables identify harvest restrictions and sampling actions for each stage of toxin concentration. When there is an increase in alert stage, the increase will be effective on the date and time that samples are collected. Alert stages below the closure alert level will be decreased effective on the sampling date and time after the sample results are received from the laboratory. To be considered for a decrease in alert stage, successive samples collected at least three (3) days apart (i.e., 72 hours) must meet the toxin level requirements for that stage.

D. SIZE OF AREA SUBJECT TO CLOSURE OR HARVEST RESTRICTION

The initial harvest closure or implementation of harvest restrictions will include the entirety of the certified areas within the SBMC's lease, unless otherwise determined in a signed biotoxin management agreement. In reopening any commercial area, or any part of that area located within two miles by water of any site where toxin in a shellfish sample was detected above the alert level, the criteria in Section G apply.

E. SPECIES RESTRICTIONS

A closure or harvest restriction of a commercial growing area may be limited to the shellfish species determined to have reached the alert level, or it may include all species in the growing area.

F. SAMPLING DURING A MARINE BIOTOXIN CLOSURE OR HARVEST RESTRICTION

Follow-up sampling will be continued throughout a closure or harvest restriction. Follow-up sampling in certified commercial shellfish growing areas may include all shellfish species that are commercially harvested in that area. CDPH will determine sampling

sites and times. Sites will be selected so that no part of any certified growing area is more than two miles by water from a sampling site for the species of shellfish harvested in that growing area.

G. REOPENING CRITERIA

Prior to being considered for reopening from a biotoxin closure, or a closure due to lack of required biotoxin samples, the Santa Barbara Mariculture Company shall collect successive samples at least three (3) days apart (i.e., 72 hours) that are found to be below the federal alert level for the toxin of concern. Reopening shall be contingent upon these sampling results. For biotoxin closures, downward trends in biotoxin levels in the surrounding area will be evaluated and considered when making a reopening decision. A reopened area is still subject to the harvest restrictions set forth in Tables 3 and 4. Reopening shall be effective on the date and time that sample results are received from the laboratory.

Harvesting during a growing area closure by a certified commercial harvester is a serious violation and may result in suspension or revocation of their growing area certificate.

XIV. GROWING AREA REEVALUATION

CDPH/PSU will review and reevaluate water quality and other sanitary survey data at least annually, in accordance with procedures set forth in the Model Ordinance, Chapter IV (2023), to confirm the classification of the growing area and to determine if changes are needed in the closure rules. At least once a year, CDPH/PSU will conduct an on-site audit to determine harvester compliance with the water quality sampling plan. The findings will be presented in an annual growing area sanitary survey update report.

When a growing area fails to meet the water quality standards for its classification, the NSSP requires that if the annual reevaluation determined that conditions have changed based on the information and data collected during the annual review and that the growing area classification is incorrect, immediate action shall be initiated to reclassify the area. If determined that an emergency condition or situation exists, then the growing area will be immediately (within 24 hours) placed in the closed status.

XV. AGREEMENT OF INVOLVED PARTIES

The NSSP Model Ordinance, Chapter IV (2023) section @.03 C.4(a), requires that "The Management Plan shall be developed by the authority, in coordination with: the local shellfish industry, the individuals responsible for the operation of any wastewater treatment plants involved and any local or State agencies." In addition, NSSP Model Ordinance, Chapter IV (2023) section @.03 C.4(b) states "Failure of any one party to agree shall constitute sufficient justification to deny creation of a Conditionally Approved area." This Management Plan is in effect as long as the area remains certified and may be modified when conditions warrant it, with the concurrence of the involved parties.

(The Sanitary Survey Report is updated annually; any significant changes in this report will necessitate a revision to the Management Plan).

A. PUBLIC AGENCIES

Authorized persons from the following public agencies are requested to review this Management Plan and to sign and return a statement of agreement provided in Appendix F, which states that "The undersigned has read and understands the purpose of the *Conditionally Approved* classification of the commercial shellfish growing area at the Santa Barbara Lease M-653-02 Area and the conditions of its Management Plan, dated _____, and agrees to comply with the conditions and procedures set forth in the Management Plan."

1. California Department of Fish and Wildlife, Wildlife Protection Division
2. California Regional Water Quality Control Board, Central Coast Region
3. Santa Barbara County Environmental Health Services

B. PERMITTED FACILITIES

Authorized persons in the following agencies are requested to review this plan and to sign and return a statement of agreement provided in Appendix F, which states that "The undersigned has read and understands the purpose of the '*Conditionally Approved*' classification of the commercial shellfish growing area at the Santa Barbara Lease M-653-02 Area and the conditions of its Management Plan, dated _____, and agrees to comply with the conditions and procedures set forth in the Management Plan."

1. City of Santa Barbara Sanitary District (alternate agreement provided due to MOU)
2. Goleta Sanitary District

C. COMMERCIAL SHELLFISH HARVESTER

Compliance with this Management Plan is mandatory for the commercial shellfish harvester as a condition of their Shellfish Growing Area Certificate issued by CDPH/PSU. Noncompliance may result in suspension or revocation of the certificate. The Santa Barbara Mariculture Company is requested to review this document and to sign and return the statement provided in Appendix F, which states that "The undersigned has read and understands the purpose of the '*Conditionally Approved*' classification of the commercial shellfish growing area at the Santa Barbara Lease M-653-02 Area and the conditions of its Management Plan, dated _____, and understands that compliance with the conditions and procedures set forth in the Management Plan is mandatory as a condition of Shellfish Growing Area Certificate issued by the California Department of Public Health."

D. FORWARDING OF SIGNED STATEMENTS

Signed statements shall be submitted to:
California Department of Public Health, EMB
850 Marina Bay Parkway, # G165, Richmond, CA 94804
shellfishpreharvest@cdph.ca.gov

XVI. SUMMARY OF AGREEMENTS

A. CALIFORNIA DEPARTMENT OF FISH AND WILDLIFE

Upon request from CDPH/PSU, to field check commercial shellfish growing areas during closed periods to monitor for illegal harvesting activity, and to report compliance monitoring activities and findings to CDPH/PSU.

B. CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD, CENTRAL COAST REGION

1. Notification

To notify, or to require designated persons or agencies to notify, CDPH/PSU of any pollution events, as soon as they become aware of discharges which may pose a threat or adversely affect water quality in the Santa Barbara coastal area.

2. Performance Standards

To include in the Waste Discharge Requirements or Monitoring and Reporting Program for all wastewater treatment plants whose wastes discharge into the Santa Barbara Channel (including the City of Santa Barbara's El Estero Wastewater Treatment Plant, and the Montecito, Goleta, Summerland, and Carpinteria sanitary districts) performance standards designed to preserve the water quality in the Santa Barbara Channel for the beneficial use of shellfish harvesting.

3. Waste Discharge Requirements

To include the following requirements in the Waste Discharge Requirements or Monitoring and Reporting Program issued to the City of Santa Barbara's El Estero Wastewater Treatment Plant and the Goleta sanitary district:

a. To notify CCRWQCB, CDPH/PSU, and the commercial shellfish harvesters as soon as possible (within 4 hours during regular business hours; by 10:00am the following morning if event occurs during off-business hours), of any sewage spill or malfunction of a wastewater treatment facility which results in a potential or actual discharge of sewage or inadequately treated effluent into the Santa Barbara coastal area.

- b. To develop and maintain written notification procedures incorporating the notification requirement as described above, to post the procedures at the facility, and to provide a copy of the current notification procedures to CCRWQCB, CDPH/PSU, and SBCEHS.
- c. To provide CDPH/PSU a copy of all Waste Discharge Requirements and updates or amendments, proposed or adopted, for any WWTPs whose wastes discharge to the Santa Barbara Channel.

C. SANTA BARBARA COUNTY ENVIRONMENTAL HEALTH SERVICES

To notify CDPH/PSU of any beach postings or closures due to high coliform levels, any sewage spills or line breaks, and of any other pollution event that they are notified of which could impact the ocean waters.

TABLES

MANAGEMENT PLAN FOR COMMERCIAL SHELLFISH GROWING AREA M-653-02
2025

Table 1. Wastewater treatment plants permitted to discharge to the Santa Barbara Channel under Waste Discharge Requirements from the California Regional Water Control Board, central coast region.

FACILITY	LOCATION	EFFLUENT (MGD)	WASTE SOURCE	TREATMENT TYPE	DISPOSAL	PERMIT EXPIRATION DATE
El Estero Water Resource Center	520 East Yanonali St, Santa Barbara, CA 93103	Max design: 18, Ave: 8	City of Santa Barbara and portions of Santa Barbara County	Screening, comminution, aerated grit removal, primary sedimentation, activated sludge, secondary sedimentation, chlorination/dechlorination	Into the Santa Barbara Channel through a 8720 ft outfall/diffuser system, approx 70' deep	January 31, 2025
Goleta Wastewater Treatment Plant	1 Moffett Place, Goleta, CA 93117	Max: 25.4, Ave (dry weather): 9.0	City of Goleta, Univ of California at Santa Barbara, Santa Barbara Municipal Airport, and facilities of Santa Barbara City	Primary sedimentation, biofiltration, solids-contact, secondary clarification, chlorination/dechlorination	Into the Santa Barbara Channel through a 5912 ft outfall/diffuser system, approx 87' deep	November 9, 2022

MANAGEMENT PLAN FOR COMMERCIAL SHELLFISH GROWING AREA M-653-02
2025

Table 2. Waste Discharge Requirements and monitoring and reporting programs for wastewater treatment plants in the Santa Barbara area.

FACILITY	TC/FC in Effluent	Max TC/FC at Growing Area	Notification Procedure	Notes
El Estero Water Resource Center	Median TC shall not exceed 2300 MPN/100 mL for last 7 days; TC in any one sample shall not exceed 16,000 MPN/100 mL. Median FC shall not exceed 460 MPN/100 mL for last 7 days; FC in any one sample shall not exceed 3200 MPN/100 mL.	Median TC shall not exceed 70 MPN/100 mL, and 90% of the samples shall not exceed a 230 MPN/100 mL for TC.	Discharger shall notify CCRWQCB, CDPH, and any shellfish harvester as soon as possible when loss of disinfection occurs or if 3 consecutive effluent total coliform bacteria tests exceed 16,000 MPN/100 mL. Plus additional plant parameter notification requirements in accordance with the MOU.	1, 2, 3
Goleta Wastewater Treatment Plant	No more than 10% of the final effluent samples in any month (30-day) period shall exceed a TC of 2400 MPN/100 mL and no sample shall exceed 16,000 MPN/100 mL.	Median TC shall not exceed 70 MPN/100 mL in any 60-day period. 90% of the samples in any 60-day period shall not exceed a 230 MPN/100 mL for TC.	If any sample exceeds 16,000 MPN/100 mL, the discharger shall notify within 48 hrs CDPH, CCRWQCB, and any certified commercial shellfish harvesters in the vicinity of the outfall.	1, 3, 4

Notes:

1. El Estero and Goleta have additional coliform limits: The discharger shall not cause the following bacteriological limits to be exceeded between the shoreline and the 30-foot depth contour/a distance of 1,000 feet from the shoreline; within areas where there are kelp beds; and within areas used for body contact recreation:

Parameter	Total Coliform (MPN/100 mL)	Fecal Coliform (MPN/100 mL)	Enterococcus (MPN/100 mL)
Geometric Mean (30 days, 5 most recent samples)	1,000	200	35
Single Sample Maximum	10,000*	400	104
*Total coliform density shall not exceed 1,000 MPN/100 mL when the FC/TC ratio exceeds 0.1			

2. For El Estero: if 3 consecutive effluent TC bacteria tests exceed 16,000 MPN/100 mL, samples shall be collected at shoreline stations and analyzed for FC and TC once a week for a minimum of 14 days until the effluent concentrations return to compliance.

Table 2. Notes Continued

3. The "Standard Provisions and Reporting Requirements of Waste Discharge Requirements", issued by the California Regional Water Quality Control Board, dated February 2004 (latest available version) contains a general requirement that "Any noncompliance that may endanger health or the environment shall be reported orally within 24 hours from the time the discharger becomes aware of the circumstances..."
4. The transport time from the Goleta outfall to the growing area under worst-case current velocity of 0.82 ft/sec is approximately eight (8) hours. For a current velocity of 0.2 ft/sec, the transport time is approximately 30 hours.

MANAGEMENT PLAN FOR COMMERCIAL SHELLFISH GROWING AREA M-653-02
2025

TABLE 3. Rainfall Closure Rules for M-653-02.

Rainfall Management Area	24-Hour Cumulative Rainfall Total	Start Closure	Closure Length = End of Storm Plus Below:
M-653-02 Lease Area	> 3.0 Inch	When the approved rain gauge exceeds threshold limit.	72-hrs (3-days)

Notes:

Start of Closure and Closure Length is reinitiated and recalculated each time 24-hour cumulative rainfall threshold is exceeded (including during periods in which closure is already in effect).

During extreme environmental conditions of greater than 5.15 inches in 24 hours, the growing area needs to be sampled to reopen, see Section VIII.A.2.

Data used in making rainfall closure thresholds is presented in the Triennial Sanitary Survey Update Report: 2023-2024 Commercial Shellfish Growing Area Lease M-653-02 Santa Barbara Channel, California.

MANAGEMENT PLAN FOR COMMERCIAL SHELLFISH GROWING AREA M-653-02
2025

Table 3. Management protocol for domoic acid in commercial shellfish growing areas.

ALERT STAGE	DOMOIC ACID CONCENTRATION	ACTION¹
No Alert	< 2.5 ppm	- Routine Sampling; - Harvest date must be < 168 hours from sampling date.
1st Stage	2.5 – 4.9 ppm	- Harvest date must be < 48 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information ² ; - Immediate resample ³ ; - Additional sampling as directed.
2nd Stage	5 – 9.9 ppm	- Harvest date must be < 24 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information ² ; - Immediate resample ³ ; Twice-weekly sampling at minimum ³ ; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Expand sampling to adjacent growing areas.
3rd Stage	10 – 19.9 ppm	- Lot release ^{3,4} ; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Immediate resample ³ ; Twice-weekly sampling at minimum ³ ; - Expand sampling to adjacent growing areas.
4th Stage	≥ 20 ppm	- Closure of growing area; - Batch release ^{3,4} of shellstock from adjacent harvest areas that remain open; - Twice-weekly sampling ³ ; - Expand sampling to adjacent growing areas.

¹ For each alert stage, the alert shall be continued until either (a) a higher stage alert is declared, or (b) two successive samples taken over a minimum period of three days have domoic acid levels consistent with a lower alert stage and downward trends in biotoxin levels in the surrounding area.

² For growing areas with more than one harvester. Notify CDPH on day of sample collection.

³ Contingent upon arrangement with certified laboratory.

⁴ Lot release means that the harvester must submit samples from each harvested lot of shellfish and shall not release that lot to market until CDPH has verified its safety based on satisfactory analytical results. Product must be held under appropriate temperature control while awaiting analytical results. Harvesters should consult the CDPH Food and Drug Branch for guidance on proper handling and storage of the batch-harvested shellstock.

Table 4. Management protocol for PSP in commercial shellfish growing areas.

ALERT STAGE	PSP TOXIN CONCENTRATION	ACTION¹
No Alert	Nondetectable or < ~34 µg/100g	- Routine Sampling; - Harvest date must be < 168 hours from sampling date.
1st Stage	Detectable²: < 50 µg/100g	- Harvest date must be < 48 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information; - Immediate resample³; - Additional sampling as directed.
2nd Stage	Detectable: < 60 µg/100g	- Harvest date must be < 24 hours from sampling date; - Notify CDPH date and time of sample collection and shipping information; - Immediate resample³; Twice-weekly sampling at minimum³; - Immediate harvest closure if harvest lot laboratory results have not been obtained; - Expand sampling to adjacent growing areas.
3rd Stage	60 - 79 µg/100 g	- Lot release^{3,4}; - Immediate harvest closure if harvest lot laboratory results have not been obtained. - Immediate resample; Twice-weekly sampling at minimum³; - Expand sampling to adjacent growing areas.
4th Stage	≥ 80 µg/100 g	- Closure of growing area; - Batch release^{3,4} of shellstock from adjacent harvest areas that remain open; - Twice-weekly sampling³; - Expand sampling to adjacent growing areas.

¹ For each alert stage, the alert shall be continued until either (a) a higher stage alert is declared, or (b) two successive samples taken over a minimum period of three days have PSP toxin levels consistent with a lower alert stage and downward trends in biotoxin levels in the surrounding area.

² A positive result for the presence of PSP toxins on a qualitative test will result in the 1st Stage Actions.

³ Contingent upon arrangement with certified laboratory.

⁴ Lot release means that the harvester must submit samples from each harvested lot of shellfish and shall not release that lot to market until CDPH has verified its safety based on satisfactory analytical results. Product must be held under appropriate temperature control while awaiting analytical results. Harvesters should consult the CDPH Food and Drug Branch for guidance on proper handling and storage of the batch-harvested shellstock.

Table 5. Company Personnel Currently Certified to Collect/Process Biotoxin Samples.

Company	Certified Sampler	Certification Date
SBMC	Bernard Friedman	January 28, 2025

FIGURES

MANAGEMENT PLAN FOR COMMERCIAL SHELLFISH GROWING AREA M-653-02
2025

Figure 1. Location of lease M-653-02, wastewater treatment plant ocean outfalls, and *Prohibited* areas in the Santa Barbara Channel.

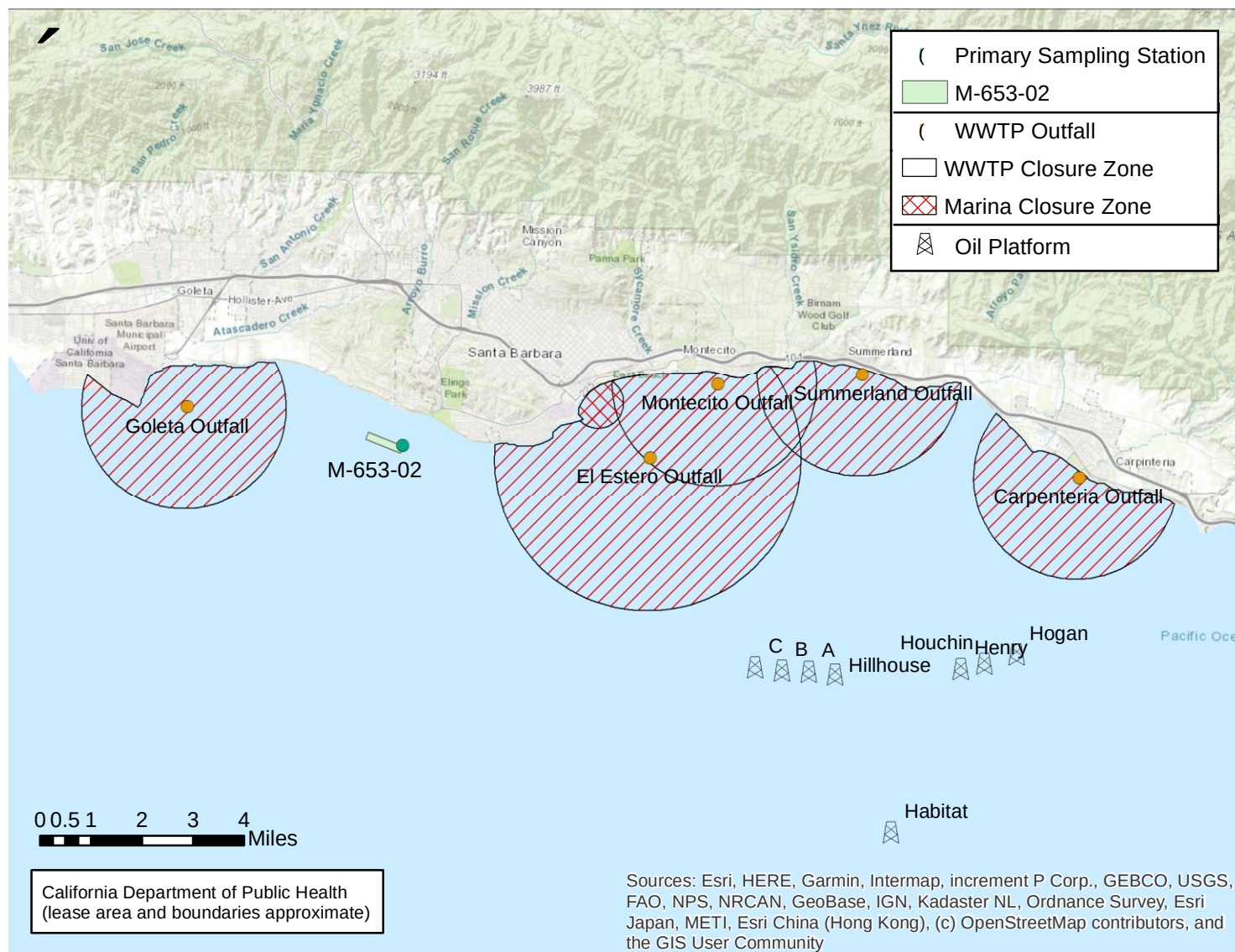
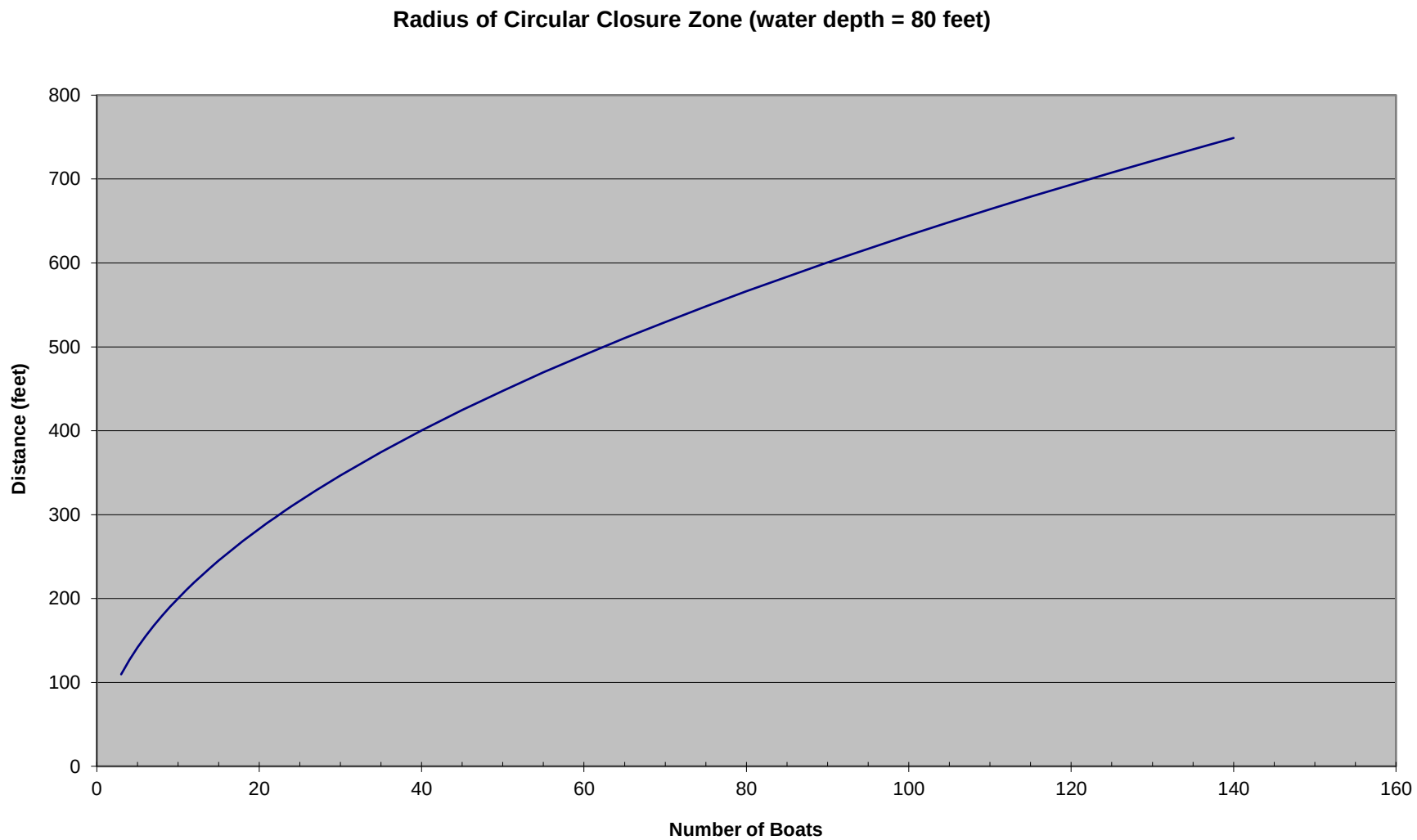


Figure 2. Calculated safe distances in Santa Barbara Channel between moored live-aboard boats and shellfish growing areas.



APPENDICES

APPENDIX A. CONTACT LIST FOR SANTA BARBARA LEASE M-653-02

CALL-DOWN CONTACT LIST

Matt Scanlon
Preharvest Shellfish Unit
California Department of Public Health
510-374-0444
Matt.Scanlon@cdph.ca.gov

Joe Christen
Preharvest Shellfish Unit
California Department of Public Health
510-750-2554
Joe.Christen@cdph.ca.gov

Steve Etter
Preharvest Shellfish Unit
California Department of Public Health
510-412-4631
Steve.Etter@cdph.ca.gov

Vanessa Zubkousky-White, Chief
Preharvest Shellfish Unit
California Department of Public Health
510-412-4635
Vanessa.Zubkousky@cdph.ca.gov

Sam Rankin
Preharvest Shellfish Unit
California Department of Public Health
510-412-4633
Sam.Rankin@cdph.ca.gov

**AFTER HOURS SHELLFISH
NOTIFICATION:**

California Emergency Management Agency
800-852-7550 or 916-262-1621

OTHER CONTACTS

Eric Trevena, Chief
Environmental Health Services Section
California Department of Public Health
Eric.Trevena@cdph.ca.gov

Bernard Friedman
Santa Barbara Mariculture Company
805-886-1283
Bernard@sbmariculture.com

Marshall Kinsey
Shellfish Specialist
U.S. Food and Drug Administration
Marshall.Kinsey@fda.hhs.gov

Christina Grant
Shellfish Program Specialist
California Department of Public Health
Christina.Grant@cdph.ca.gov

Kristina Phelps
Shellfish Specialist
U.S. Food and Drug Administration
Kristina.Phelps@fda.hhs.gov

Sara Briley
Marine Aquaculture Coordinator
California Dept. of Fish and Wildlife
Sara.Briley@Wildlife.ca.gov

Randy Lovell
Aquaculture Coordinator
California Dept. of Fish and Wildlife
Randy.Lovell@wildlife.ca.gov

Thomas Welche
Wastewater Treatment Superintendent
El Estero Wastewater Treatment Plant
TWelche@SantaBarbaraCA.gov

Steve Wagner
General Manager
Goleta Sanitary District
SWagner@goletasanitary.org

Marco Felix
Chief Plant Operator
Montecito Sanitary District
MFelix@montsan.org

Craig Murray
General Manager
Carpinteria Sanitary District
CraigM@carpsan.com

David Lewis
Operations Manager
Summerland Sanitary District
NAguilarVega@summerlandsd.org

Rebecca Bjork
Water Resources Manager
City of Santa Barbara, Public Works
Dept.
RBjork@SantaBarbaraCA.gov

Katherine Faick, Chief
Ocean Standards Unit
State Water Resources Control Board
Katherine.Faick@Waterboards.ca.gov

Cassidy Teufel
California Coastal Commission
Cassidy.Teufel@coastal.ca.gov

Central Coast Regional Water Quality
Control Board
centralcoast@waterboards.ca.gov

Tina Yin
U.S. Environmental Protection Agency
Yin.Tina@epa.gov

APPENDIX B. MEMORANDUM OF UNDERSTANDING

MEMORANDUM OF UNDERSTANDING
BY AND BETWEEN
THE
CALIFORNIA DEPARTMENT OF HEALTH SERVICES
AND
CITY OF SANTA BARBARA

This Memorandum of Understanding (hereinafter MOU), is made and entered into by and between the California Department of Health Services (hereinafter DHS), and the City of Santa Barbara (hereinafter City).

The purpose of this MOU is to facilitate DHS's ability to manage shellfish growers in the Santa Barbara Channel in conformance with the standards and related procedures of the National Shellfish Sanitation Program (hereinafter NSSP).

WHEREAS, the U.S. Food and Drug Administration administers the NSSP; AND

WHEREAS, California statute, contained in the Health and Safety Code, requires DHS to develop and administer a shellfish sanitation and control program which conforms to the standards and related procedures of the NSSP; AND

WHEREAS, the NSSP standards and related procedures are set forth in the NSSP Model Ordinance (1997 Revision); AND

WHEREAS, the NSSP Model Ordinance provides the definitions, criteria, and requirements for classification of shellfish growing areas; AND

WHEREAS, the NSSP Model Ordinance definition, criteria, and requirements for conditional classifications of shellfish growing areas include: 1) the development of management plans based on wastewater treatment plant function and performance standards; and 2) procedures for immediate notification when wastewater treatment plant performance standards and criteria are not met; AND

WHEREAS, the City owns and operates the El Estero Wastewater Treatment Plant (hereinafter EEWWTP); AND

WHEREAS, the EEWWTP effluent outfall is located 1½ miles offshore of the Santa Barbara coast; AND

WHEREAS, the California Regional Water Quality Control Board (hereinafter CRWQCB), on September 8, 1999, adopted Waste Discharge Requirements (hereinafter WDR) Order No. 99-40 (National Pollutant Discharge Elimination System Permit No. CA0048143) for EEWWTP; AND

WHEREAS, the WDR's were adopted in part to protect beneficial uses of the Pacific Ocean, including shellfish harvesting; AND

WHEREAS, the effluent of the EEWWTP may potentially impact the water quality of nearby shellfish growers if there is a loss of effluent disinfection; AND

WHEREAS, Monitoring and Reporting Program No. 99-40 of the WDR's requires "Discharger shall notify the CRWQCB (telephone: 805 549-3147), DHS (telephone: 510-540-3423), and any Mariculture Grower as soon as possible when there is a loss of disinfection or if three consecutive effluent coliform bacteria tests exceed 16,000 MPN/100ml."; AND

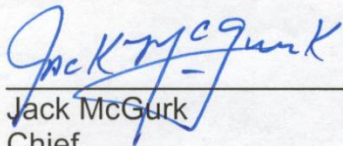
WHEREAS, the City has developed and implemented the "Operator Effluent Monitoring and Early Warning Notification Procedure" at EEWWTP in accordance with the WDR's and with the concurrence of DHS as a further measure to protect the environment and public health. [A copy of the "Operator Effluent Monitoring and Early Warning Notification Procedure" (dated July 19, 2000), is attached and made a part of this MOU.]; AND

WHEREAS, both the City and DHS have an interest in protecting the environment and public health.

NOW, THEREFORE, it is mutually agreed and understood as follows:

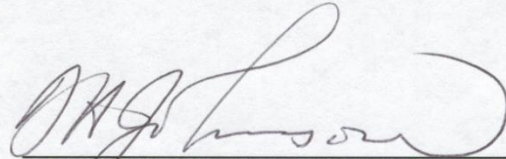
1. The City and DHS have worked together in establishing a routine Monitoring and Early Warning Program for the purpose of providing adequate levels of safety for the shellfish growing waters through the evaluation, acceptance and implementation of the "Operator Effluent Monitoring and Early Warning Notification Procedure."

2. This MOU and the "Operator Effluent Monitoring and Early Warning Notification Procedure", may be amended as mutually agreed upon by the City and DHS.
3. The effective date of this MOU shall be the last date signed below.
4. This MOU can be terminated unilaterally by the City with 60 days written notice.



Jack McGurk
Chief
Environmental Management Branch
Department of Health Services
State of California

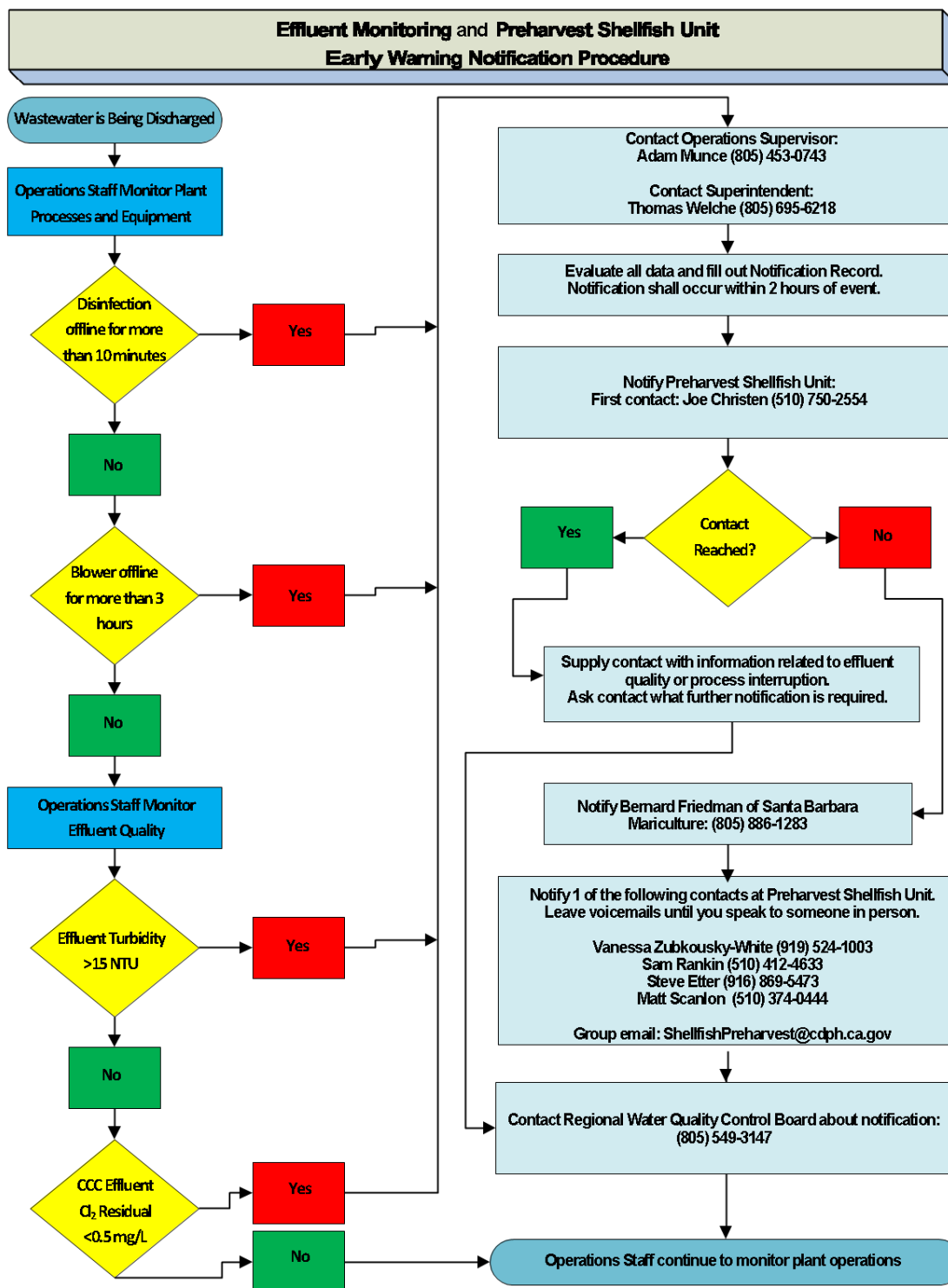
01-31-01
Date



David H. Johnson
Director
Public Works Department
City of Santa Barbara

01-23-01
Date

Figure B-1. El Estero Effluent Monitoring and Notification Procedure



APPENDIX C. WASTEWATER UPSET RESPONSE PROCEDURES

The following procedures shall be followed in the event of a failure in the wastewater collection or treatment system resulting in a “wastewater upset”. A wastewater upset occurs whenever inadequately treated wastewater has exited the confines of the wastewater treatment or collection system and has the potential to enter the Santa Barbara Channel.

In the event of a wastewater upset that results in a potential or actual discharge of inadequately treated effluent into the Channel, the responsible agency will notify CDPH and the harvester of the location and estimated volume of wastewater released to the waterbody. Upon notification of a sewage spill SBMC shall consider the growing area closed to harvesting. CDPH/PSU will contact SBMC to confirm that notification of the wastewater upset event has been received, and that the harvester has ceased harvest operations after the start of the upset. Harvesting shall not occur until after CDPH/PSU has evaluated the event and determined the required actions. If harvesting has occurred after the start of the spill, the harvester should consult CDPH Food and Drug Branch to determine the disposition of harvested shellstock.

Spill Volume Thresholds

CDPH/PSU will establish that the spill has been abated and determine if spill volume and location warrant closure of the growing area. Minimum spill volume thresholds for raw sewage that will result in closure are presented in Table C-1. For spill locations not specified in Table C-1, thresholds will be determined by interpolation or extrapolation from a nearby location in Table C-1. CDPH/PSU may modify the above procedure based on the specific conditions of the reported spill. After the determination has been made, CDPH/PSU will notify the harvester of the status of the growing area.

Spill Attenuation

CDPH/PSU will conduct analyses to estimate when fecal coliform concentrations resulting from an upset will be attenuated to acceptable levels (14 MPN/100 mL). Attenuation analysis will be conducted as described in the Sanitary Survey for Commercial Shellfish Growing Area Lease M-653-02. The analysis accounts only for dispersion of the sewage spill by currents, and ignores bacterial die-off. This conservative approach is appropriate because die-off rates of many of the pathogenic viruses associated with sewage, and for which fecal coliforms serve as imperfect indicators, are unknown.

CDPH/PSU will instruct SBMC to begin collecting samples when the attenuation analysis indicates acceptable water quality. The harvester should not sample until instructed to do so by CDPH.

Sampling and Opening Criteria

The sample to reopen requirements below may be removed 21 days (504 hours) after the end of spill or upset.

Sampling to reopen a closed area after a sewage spill is contingent upon identification and availability of an appropriate FDA-certified laboratory that can perform male-specific coliphage

(MSC) analysis for shellfish meat.¹ If a laboratory is unavailable, the closed portions of the growing area will remain closed for at least 21 days (504 hours) after the end of the spill.

If a laboratory is available, and as directed by CDPH, harvester(s) shall collect whole in-shell shellfish sample(s) from beds closest to the spill input location for MSC analysis at the harvester's expense. CDPH/PSU may require samples from multiple representative areas of the bay. Samples can be collected starting at a date and time no sooner than 7 days (168 hours) after the spill has ended.² The sampling requirement ends after 21 days have passed since the end of the spill.

All representative shellfish meat samples must be equal to or less than 50 plaque forming units per 100 grams (PFU/100 g) MSC to be considered for reopening. If any shellfish meat sample exceeds 50 PFU/100 g, harvesters will be instructed to submit an additional sample from the same bed groups. The reopening date and time of the entire affected area shall be the date and time CDPH/PSU staff receive and have reviewed the last of all acceptable results from the laboratory. Partial reopening will not be considered.

Table C-1. Raw Sewage Upset Threshold Volumes

Waterway or Outfall	Raw Sewage* Threshold (gallons)	Undisinfected Effluent** Threshold (gallons)	Distance to M-653-02 (miles)	Average Depth (feet)
Arroyo Burro	87,401		0.8	45
Goleta Outfall	3,736,011	37,360,110	3.6	88
Goleta Slough	2,276,795		3.9	45
Mission Creek	2,288,366		3.9	45
El Estero Outfall	5,789,036	57,890,360	4.7	80
Sycamore Creek	3,789,212		5.1	45
Montecito Outfall	8,061,082	80,610,820	6.3	62
Devereux Slough	6,816,334		6.8	45
Summerland Outfall	14,593,795	145,937,950	9.0	55
Carpinteria Outfall	42,074,680	420,746,800	13.0	57

* FC concentration of 1.4×10^7 MPN/100 mL.

** FC concentration of 1.4×10^6 MPN/100 mL for secondary treated.

¹ See Domestic NSSP Laboratory List: <https://www.issc.org/laboratory-1>

² NSSP Model Ordinance (2023)

Figure C-1. Wastewater upset decision tree for California Department of Public Health.

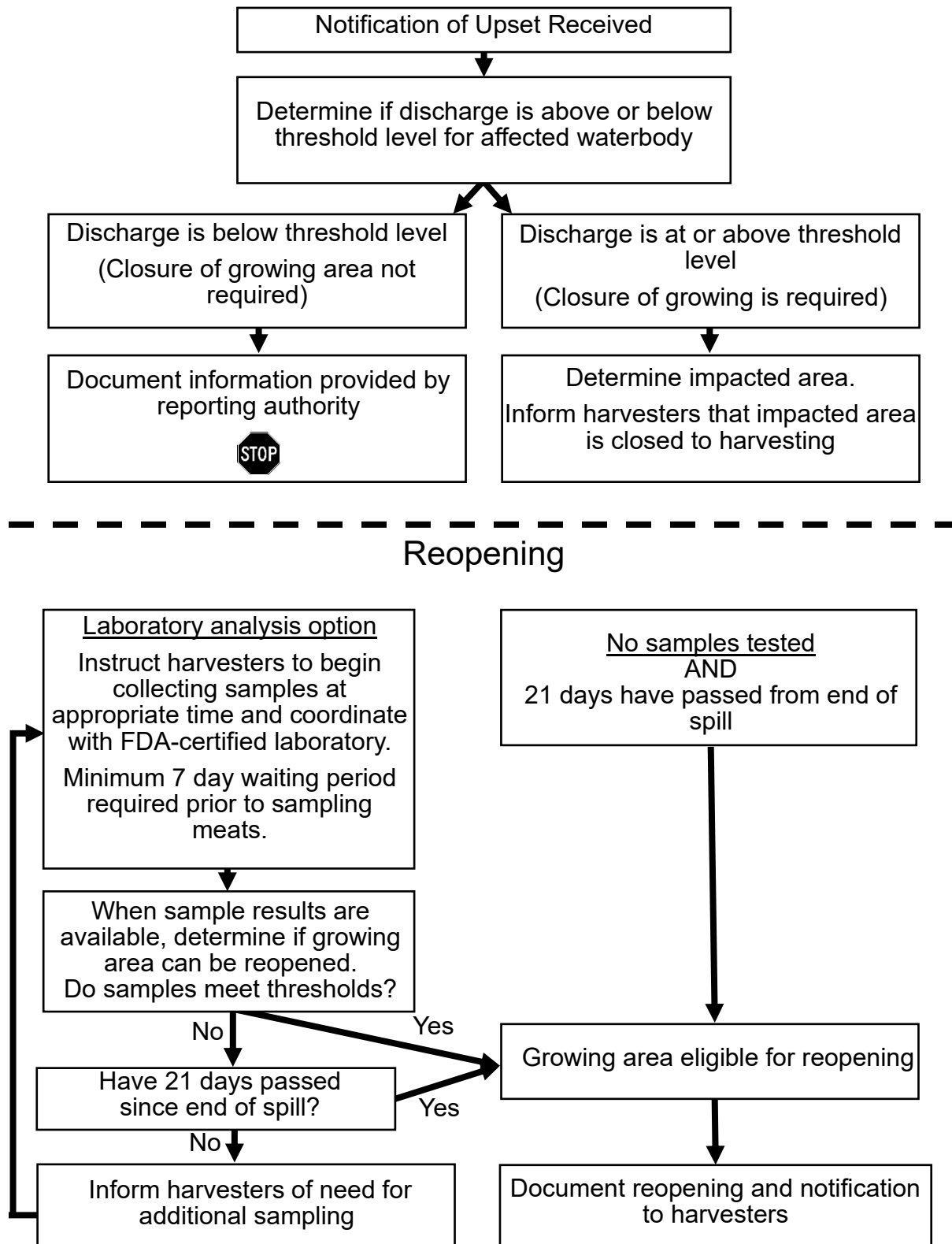
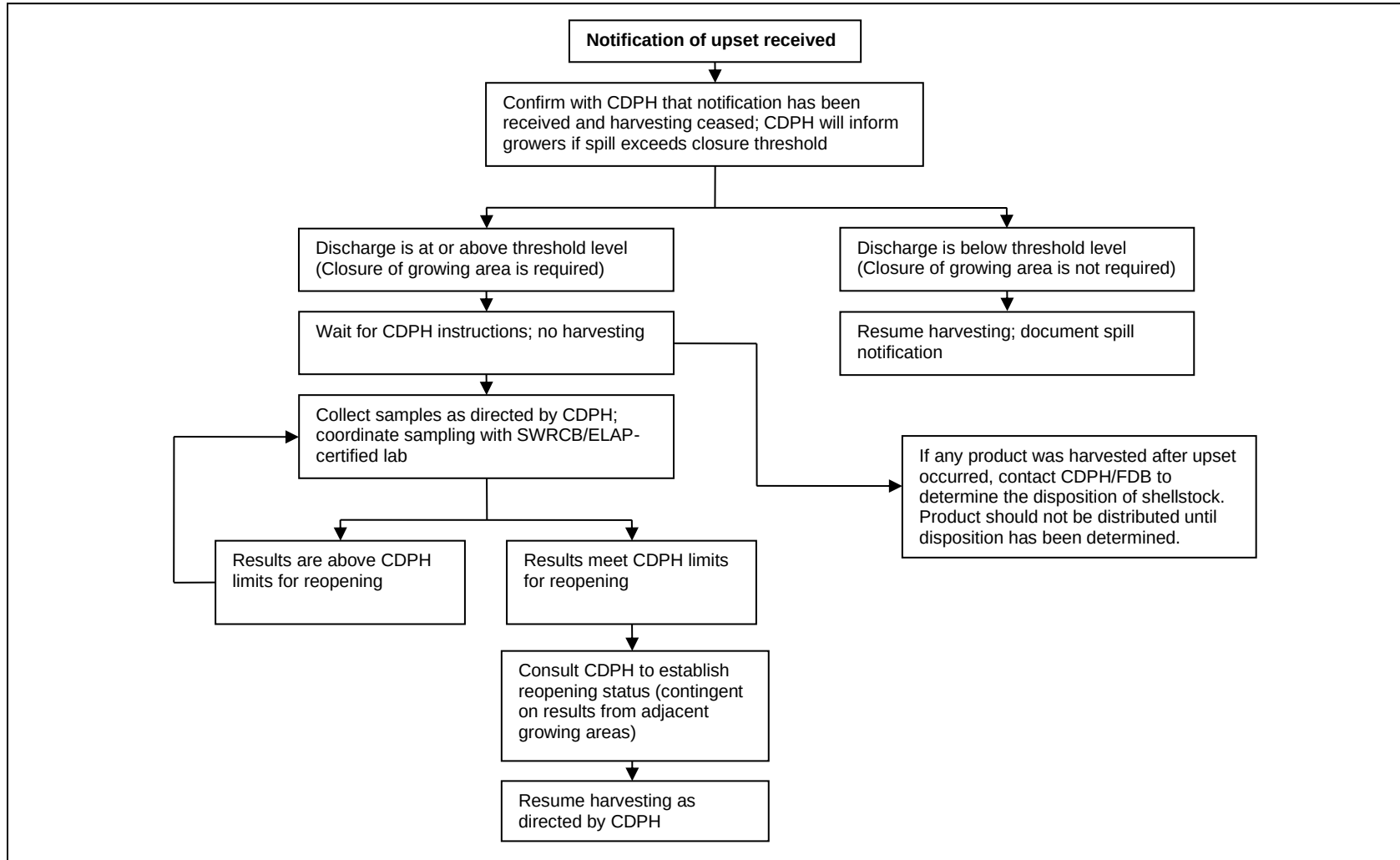


Figure C-2. Wastewater upset decision tree for harvesters.



APPENDIX D. SHELLFISH HARVESTER'S MONTHLY REPORT OF HARVEST
OPERATIONS

HARVESTER _____
GROWING AREA _____

MONTH/YEAR _____
RAIN GAUGE LOCATION _____

This form is for the use of California certified shellfish harvesters who harvest from conditionally approved areas. Harvesters are required to complete and file this form in accordance with the Management Plan for the conditionally approved area in which they conduct harvesting operations.

Day	Closure & Reopening Dates and Times	Harvest Times & Amounts
1		
2		
3		
4		
5		
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7		
8		
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10		
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APPENDIX E. MARINE BIOTOXIN MONITORING

This appendix contains two protocols for marine biotoxin monitoring. Procedure 1 is to be used for collecting shellfish samples to be submitted to the California Department of Public Health for biotoxin analysis. Procedure 2 should be used when collecting phytoplankton samples. Training in both procedures will be provided to the harvester by CDPH.

PROCEDURE 1: COLLECTING SHELLFISH SAMPLES TO BE SENT TO CDPH FOR BIOTOXIN ANALYSIS (NOT ANALYZED IN THE FIELD)

Prepared By

California Department of Public Health
Environmental Management Branch and Environmental Microbial Diseases
Laboratory

INTRODUCTION

The following field sampling protocol will accommodate analysis for both paralytic shellfish poisoning (PSP) toxins and domoic acid (DA). Because a preservative cannot be used at the time of sample collection, it is imperative that the field collectors take care to ensure the integrity of each sample. Regulatory growing area samples must meet the National Shellfish Sanitation Program standards for sample handling and timing. Regulatory samples must be processed (homogenized) by the laboratory within 24 hours of collection. The regulatory sample time limit can be extended if the shucked sample is frozen. If a shucked sample is not frozen it must be collected in the afternoon, adequately chilled, and shipped to arrive the following morning by 10:30 AM.

1. **SAMPLING:** A sample should consist of a single species of bivalve shellfish (mussels, oysters, clams, etc.) collected randomly from the sampling site. For harvested species, the sample should be representative of harvested product. Each sample should include a **minimum of 20 individuals and at least 250 grams of drained shellfish meat**; this provides adequate material for both analyses with a reserve as insurance against a possible lab accident. This is equivalent to a volume of about one (1) cup of shucked meats. It takes up to 40 small sea mussels (about 2 inches shell length) to produce 250 grams of meat. Avoid collecting only a few very large specimens: this may provide misleading information on the presence or absence of toxin. Immediately after collection the sample must be placed in dry storage (cooler) with ice or cold packs until shucking.
2. **FIELD PREPARATION OF SAMPLES:**
 - a. Shucked samples.
 - (1) Thoroughly clean the outside of shellfish with fresh water.
 - (2) Open shell by cutting the adductor muscle(s). Do not use heat or anesthetic before opening shell. **Cut carefully to avoid damage to body of mollusk.**
 - (3) Rinse the opened shellfish with fresh water to remove sand or other foreign materials.
 - (4) For mussels, cut off byssal threads (attachment hairs) with scissors and discard, saving all meat.
 - (5) Open shell and drain off excess liquid. Remove meat from shell without damaging tissue.
 - (6) Drain shucked meat on a #10 mesh sieve without layering for five (5) minutes.

- (7) Place drained meat into a wide-mouthed, 16-ounce sample bottle. About 1/2 to 2/3 of a sample bottle of shellfish meat provides the desired amount. Do not overfill; be sure to leave an air space to accommodate expansion upon freezing.

NOTE: a Ziploc® bag may be used only if a sample bottle is not available. Double or triple-bag to prevent leakage.

- (8) Tighten cap securely. Refrigerate immediately in the field.

- (9) **Freeze sample** within six hours; ensure that sample is frozen prior to shipment.

If the sample is not frozen, it must be chilled and maintained below 10 °C and arrive at the laboratory within 22 hours and before 10:30 AM to allow for processing time. If the sample temperature is greater than 10 °C, it will be rejected.

- (10) Fill out sample submission slip; be sure to record sample **bottle number** and **type of shellfish** (e.g., Pacific oyster, bay mussel, etc.). A list of representative sample types and their associated codes is presented in Table 1. In addition, please record the **sample location, sampling date and time, the collector and processor (shucker) name, shucking date and time, the date and time the sample was placed in the freezer, and include a telephone number** so that we may contact you immediately. The presence of high toxin concentrations may necessitate harvest closure and immediate resampling.

- b. Unshucked samples: If you have been instructed or authorized to send unshucked live shellfish samples, clean the outside of shellfish, wrap in newspaper to cover sharp edges, place in a heavy gauge plastic bag and seal, double bag the sample, and check for leaks. Place in dry cold storage (cooler or refrigerator) immediately. Temperature must be maintained between 0 and 10 °C, and cannot be frozen. Unshucked samples must arrive to the lab at the arraigned time and within 22 hours of collection to allow time for processing.

3. **SHIPPING:** Rapid testing of samples for biotoxins is extremely important. Samples should be shipped as soon as possible by the most rapid means available, while taking care that they arrive at the laboratory within the time and temperature requirements (< 22 hours and < 10 °C or frozen). Regulatory samples arriving outside of the proper holding temperature and timeline will be rejected, requiring immediate resampling and possible harvest closure.

- a. Place sample(s) in a sturdy insulated shipping container with an adequate quantity of frozen ice packs and sandwiched in absorbent materials to soak up any leakage or condensation.
- b. Place sample submission slip(s) on top of the Styrofoam lid; close and seal the shipping container.

- c. Containers returned to you from the laboratory will have mailing labels inside a plastic mailing envelope taped onto the lid or side. Remove the label addressed to you and reverse labels so that the box is addressed to the laboratory. Remove or cover any old shipping label(s) that could cause confusion.
- d. Package should be addressed to:
California Department of Public Health
MDL/EMDS
ATTN: Specimen Receiving
850 Marina Bay Parkway
Richmond, CA 94804
- e. Send package by Courier Service.
 - (1) You are required to make arrangements with a courier at your own expense.
 - (2) Packaging materials may be provided by CDPH. It is ultimately the harvester's responsibility to provide proper shipping materials for sample transport.
- f. Do not sending samples at the end of the week or just before holidays. Prolonged transit time causes spoilage and rejection of the sample by the laboratory.
- g. All questions regarding this protocol should be directed to the CDPH/PSU office at (510) 412-4635.

Table 1. Common sample types and their respective codes.

SAMPLE TYPE	CODE
Wild Sea Mussels (<i>Mytilus californianus</i>)	MUSW
Sentinel Sea Mussels	MUSS
Wild Bay Mussels (<i>Mytilus galloprovincialis</i>)	MUBW
Sentinel Bay Mussels	MUBS
Cultured Bay Mussels	MUBC
Cultured Pacific Oysters (<i>Crassostrea gigas</i>)	OYPC

PROCEDURE 2: PHYTOPLANKTON SAMPLING

Collecting Phytoplankton Sample

Designated commercial harvesters have been provided with a 20 µm mesh plankton net fitted with a 200 mL collection bottle at the cod end. Please follow the following guidelines as closely as possible to ensure the comparability of all samples. Obtain as dense a sample as reasonably possible.

1. Use plankton net to obtain 300 mL sample: retrieve net very slowly.
 - a. Vertical Tows: Perform vertical tows whenever possible. Many species of phytoplankton can migrate a surprising distance vertically, thus vertical tows are more likely to adequately sample these organisms. Standard tow depths are 50 feet; in shallower water you should sample from approximately 1 foot above the bottom. In extremely shallow water (< 5 feet) you can conduct horizontal tows (see 3.b.). A good rule of thumb is that three to five vertical hauls of the net should produce an adequate sample. Keep in mind that the objective is to obtain a dense sample: the color of water in the net and the rate at which the plankton net drains will provide you with an indication of the sample density.
 - b. Horizontal Tows: Whenever it is impractical to collect a phytoplankton sample with a vertical tow (e.g., in extremely shallow water) you may use this method. Slowly pull the net horizontally, just below the water surface, either along a pier or behind a boat. Never pull the net behind a boat while under way! The fine mesh net will easily tear under such stress. A drifting boat will provide enough movement to keep the net moving at the surface. When sampling from a boat try to keep track of the distance covered or the elapsed time of the tow so that you can be consistent each time you sample.
2. Record total length of the tow (depth x number of repetitions, e.g. 10 feet x 5 tows = 50 feet).
3. Swirl net sample to resuspend and fill 125 mL bottle containing 1 mL formalin just to the shoulder. If sending sample to CDPH the next day, a live sample without formalin may be submitted.
4. Pour the remaining sample into a clean bottle without formalin for field observation.

Field Notes

Please keep complete records for all samples. The laboratory submission form inside the sample shipping canister should be filled out completely.

- a. Bottle #: This is the number on the cap of each sample bottle.
- b. Tow Depth: This is the maximum depth sampled with a vertical tow (see item 3.a.).

- c. Tow Length: This is the total distance that the net was pulled to obtain a sample. For vertical tows this is equivalent to the tow depth multiplied by the number of net retrievals. For horizontal tows you can estimate the distance you walk along a pier; horizontal tows from a boat should be approximated to the best of your abilities.
- d. Whenever possible record the temperature and salinity of the surface waters. Temperatures can be obtained by collecting a bucket of water and placing a thermometer in it.
- e. Record your observations of water color, atmospheric conditions, etc. These observations can be very useful in interpreting the data.

Observing Phytoplankton Sample

Training on phytoplankton identification, including ID forms and guides, can be provided by CDPH. Below are brief guidelines to follow. Please contact the CDPH/PSU office or shellfish specialist for additional information.

1. Allow live sample to settle for at least 15 minutes.
2. Pipette several drops of the settle material to a microscope slide and place a coverslip on the drops, or use a small flat glass capillary tube to pull up a quantity of the settle material.
3. Observe the slide/capillary for the presence of *Pseudo-nitzschia*, *Alexandrium*, and *Dinophysis* and record estimate of percent composition or relative abundance (Absent, Rare < 1%, Present 1-9%, Common 10-49%, Abundant > 50%).
4. Note other most common species along with an estimate of percent composition or abundance.
5. Fill out data sheet with observations and percentages.
6. Repeat steps 6 through 9 for a total of three separate observations.
7. Field sheets should be copied before shipping to CDPH.

Documentation

Make copies of all field sheets and retain copies; ship originals as described below. Field identification sheets can also be emailed to redtide@cdph.ca.gov.

Shipping Samples

1. Place plankton field sample bottle in a Ziploc bag. Place plankton field sheet in another Ziploc bag. Enclose both in a third bag and label "For EMB".
2. Put the bagged phytoplankton sample and the 500 mL bottle of shucked shellfish in a shipping box with blue ice for shipment to CDPH via Next Business Morning delivery.

Maintenance of Net.

Please rinse the plankton net, collection bucket, and all connectors thoroughly with freshwater after each use. It is also advisable to hang-up the net to air-dry after washing out of direct sunlight. Clean with washing soda if needed.

APPENDIX F. STATEMENTS OF AGREEMENT

STATEMENT OF AGREEMENT FOR PUBLIC AGENCIES

The undersigned has read and understands the purpose of the *Conditionally Approved* classification of the commercial shellfish growing area at the Santa Barbara Lease M-653-02 Area and the conditions of its Management Plan, dated February 2025, and agrees to comply with the conditions and procedures set forth in the Management Plan

Signature

Name (print or type)

Agency

Date

Submit completed form to:

California Department of Public Health, EMB
850 Marina Bay Parkway, # G165
Richmond, CA 94804
shellfishpreharvest@cdph.ca.gov

STATEMENT OF AGREEMENT FOR HARVESTER

The undersigned has read and understands the purpose of the *Conditionally Approved* classification of the commercial shellfish growing area at the Santa Barbara Lease M-653-02 Area and the conditions of its Management Plan, dated February 2025, and understands that compliance with the conditions and procedures set forth in the Management Plan is mandatory as a condition of the Shellfish Growing Area Certification issued by the California Department of Public Health.

Signature

Name (print or type)

Agency

Date

Submit completed form to:

California Department of Public Health, EMB
850 Marina Bay Parkway, # G165
Richmond, CA 94804
shellfishpreharvest@cdph.ca.gov

APPENDIX G. RECORD OF RECENT CHANGES

TECHNICAL REPORT #	DATE	CHANGE
20-02	1/8/2020	Changed Section X.I. Wildlife to Aquaculture Operations.
20-04	1/12/2021	Modified Section X. Certified Shellfish Harvester Responsibilities.
22-04	1/27/2022	Changed the maximum seed size for Kumamoto, Pacific, and other oyster.
22-04	1/27/2022	Added meat sampling requirement for reopening from raw sewage closures.
23-04	12/19/2022	Removed Executive Summary Section. Renumbered Sections.
24-04	2/13/2024	Added Transplant Operational Plan to IX.G.2
24-04	2/13/2024	Added <i>Dinophysis</i> thresholds to XIII.A.4
24-04	2/13/2024	Removed SOP template appendix
24-04	2/13/2024	Removed 'raw' from Wastewater Response Appendix, so raw and partially treated sewage are treated the same in conformance with the unpublished 2023 Model Ordinance.
24-32	11/22/2024	Removed the Montecito, Carpinteria, and Summerland WWTP operators as lease is outside of their zones of influence.
24-32	11/22/2024	Adopted new sewage spill volume thresholds in Table C-1
24-32	11/22/2024	Implemented a 3-day closure interval for rainfall events of greater than 3 inches in 24 hours and less than 5.15 inches in 24 hours. Removed the sample to reopen requirement for FC following rainfall events of 5.15 inches or less in 24 hours.
25-07	2/14/2025	Minor language clarifying responsibilities in XIII Marine Biotxin Monitoring. Changed all references to NSSP to the newest version (2023). Changed primary CDPH contact to Matt Scanlon.